Monetary and Macroprudential Policies in a Bubbly Small Open Economy

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April 2026

Abstract. This paper investigates the optimal coordination of monetary and macroprudential policies in a small open economy exposed to global financial volatility. We develop a Dynamic New Keynesian (DNK) model featuring heterogeneous, credit-constrained entrepreneurs who trade a rational asset bubble for liquidity, alongside financial intermediaries subject to reserve requirements. We show that asset bubbles act as a powerful multiplier for exogenous disturbances. Specifically, unexpected foreign interest rate hikes burst domestic bubbles, depleting corporate collateral and severely contracting aggregate investment. Through the welfare evaluation, we demonstrate that utilizing the standard monetary policy rate to respond to asset prices—leaning against the wind—is strictly suboptimal in an open economy, as it exacerbates exchange rate volatility and imported inflation. Instead, a coordinated regime that pairs a standard Taylor rule for nominal stability with active, countercyclical macroprudential reserve requirements successfully insulates the domestic credit cycle and maximizes household welfare.

Keywords: Asset bubbles, Monetary policy, Macroprudential policy, Small open economy, Liquidity, Multiple equilibria, DSGE modeling

1. Introduction

Emerging and developing economies are increasingly exposed to fragmentation risks, such as global trade tensions, supply chain disruptions, and financial decoupling. In an era of sustained uncertainty, global shocks are more frequent and global in scope, threatening macroeconomic stability and rendering conventional policy frameworks less effective. This environment calls for a re-examination of policy designs, particularly concerning monetary policy transmission, capital flow volatility, and financial fragmentation. As capital flows become more volatile due to external shocks, the interaction between domestic asset price bubbles and central bank frameworks becomes a critical determinant of economic resilience. This paper investigates how a small open economy can optimize the design and coordination of monetary and macroprudential policies to mitigate financial fragmentation risks, manage capital flow volatility, and promote sustainable growth.

To explore these issues, we develop a Dynamic New Keynesian (DNK) model for a small open economy that incorporates rational asset bubbles and financial frictions. In our framework, heterogeneous, credit-constrained entrepreneurs face idiosyncratic investment efficiency shocks and can hold an intrinsically useless, recurrent bubble asset to store net worth and provide liquidity. Domestic banks intermediate both domestic savings and foreign capital, subject to reserve requirements that the central bank can adjust macroprudentially. We first conduct a steady-state analysis to examine the conditions under which a recurrent asset bubble can emerge. This involves characterizing both bubbleless and bubbly equilibria and analytically evaluating how domestic policy parameters—such as inflation targets and reserve requirements—and external financial conditions shape the economy's investment efficiency thresholds.

In the steady-state analysis, we find that the emergence and size of asset bubbles are highly sensitive to both global financial conditions and domestic policy stances. A looser external financial environment, characterized by lower foreign interest rates, induces capital inflows and depresses domestic real interest rates, thereby fueling the formation of asset bubbles. Conversely, tightening macroprudential policy via higher reserve requirements absorbs a larger share of aggregate net worth, displacing entrepreneurial demand for bubble assets and hindering their formation. While bubbles can crowd out efficient investment on the extensive margin by widening credit spreads, they provide crucial internal funding on the intensive margin, ultimately presenting complex trade-offs for capital accumulation and long-run output.

Building on the steady-state framework, we conduct a business cycle analysis to evaluate the dynamic responses of the small open economy to various macroeconomic disturbances. We examine three distinct shocks: a domestic short-run productivity shock, an exogenous foreign interest rate shock reflecting global financial tightening, and a non-fundamental sentiment bubble shock. Crucially, we assess the welfare implications and stabilization effectiveness of alternative policy regimes. These include a standard Taylor Rule (TR), an Augmented Taylor Rule (ATR) where monetary policy leans against asset price fluctuations, and a coordinated Macroprudential Policy

(MACRO) regime that utilizes countercyclical reserve requirements to manage credit supply.

Our dynamic analysis reveals that asset bubbles significantly amplify macroeconomic fluctuations through a "bubble multiplier" effect, particularly in response to foreign interest rate shocks. When global financial conditions tighten, capital outflows and currency depreciation trigger a collapse in bubble valuations, which depletes entrepreneurial net worth and severely contracts aggregate investment. We find that utilizing the policy interest rate to directly target asset prices (the ATR regime) backfires in an open economy setting, as it exacerbates currency depreciation and imported inflation. In contrast, deploying macroprudential tools to directly target the financial friction—by lowering reserve requirements during a bust—successfully insulates the domestic credit cycle. This coordinated MACRO regime achieves superior welfare outcomes by stabilizing asset prices and investment without compromising the central bank's nominal stability objectives.

Related literature. Our paper is closely related to four main strands of literature. First, it is related to the literature on rational asset bubbles in infinite-horizon models featuring financial frictions. A central insight of this literature is that when entrepreneurs face uninsurable idiosyncratic investment shocks and credit constraints, intrinsically useless assets can command a liquidity premium by serving as a store of value and relaxing future borrowing limits (Kocherlakota 2009; Miao and Wang 2018). We build directly on this foundation, utilizing the continuous emergence and stochastic bursting of asset bubbles to generate a "bubble multiplier" effect. Our paper closely aligns with Dong, Miao, and Wang (2020) and Dong, Mei, and Xiao (2025), who demonstrate that asset bubbles can survive the transversality condition in Dynamic New Keynesian (DNK) frameworks when collateral constraints bind, fundamentally altering the economy's steady-state capital allocation and intensive investment margins.

Second, our work contributes to the extensive debate on whether monetary policy should respond to asset price inflation, commonly referred to as the "leaning-against-the-wind" policy. Following the foundational arguments of Bernanke and Gertler (1999) that central banks should focus strictly on inflation and the output gap, subsequent theoretical work has been deeply divided. While Galí (2014) challenges the efficacy of raising interest rates to curb bubbles in overlapping-generations models, more recent infinite-horizon DNK models suggest that responding to asset prices can yield welfare improvements under specific conditions (Dong, Miao, and Wang 2020; Galí 2021; Dong, Jiao, and Wang 2025). We contribute to this debate by demonstrating that while leaning against the wind may be optimal in a closed economy, deploying the standard policy rate to target asset prices in a small open economy exacerbates exchange rate volatility and imported inflation.

Third, our work contributes to the literature that studies the intersection of asset bubbles, financial intermediary frictions, and macroprudential policy. The collapse of asset bubbles severely impairs the balance sheets of financial intermediaries, leading to spikes in credit risk premiums and sharp contractions in credit supply—a dynamic that activates the classic financial accelerator Gertler and Karadi (2011). Dong, Mei, and Xiao (2025) illustrate how asset bubbles and banking

leverage constraints create a dual amplification mechanism that propagates economic downturns. We extend this literature by explicitly modeling the central bank's reserve requirement as an active macroprudential tool, showing that countercyclical reserve adjustments can surgically address financial sector frictions without disrupting the central bank's primary nominal anchors.

Finally, our framework intersects with the literature on open-economy macroeconomics and the international transmission of financial shocks. Emerging economies are highly vulnerable to global financial cycles and foreign interest rate fluctuations, which routinely trigger sudden stops, capital flow reversals, and asset price volatility (Mendoza 2010; Miranda-Agrippino and Rey 2020). Recent advancements have integrated asset bubbles into this open-economy context, demonstrating that foreign interest rate shocks exert a substantially larger impact on domestic investment and output than domestic productivity shocks, primarily by starving the domestic bubble of liquidity (Miao, Wang, and Zhou 2022; Dong, Jiao, and Wang 2025). Our model complements this strand by evaluating the normative policy implications of global fragmentation, proving that a coordinated framework utilizing localized macroprudential tools provides superior insulation against global financial tightening compared to standard monetary policy adjustments.

Organization of the paper. Section 2 establishes the model. Section 3 examines the conditions for the existence of bubbleless and bubbly equilibria. Section 4 presents the impulse responses to shocks. Section 5 discusses the welfare implications of alternative monetary and macroprudential policy regimes. Section 7 concludes the paper. Appendices include all technical proofs and supplement materials.

2. The Model

We consider a small open economy populated by a representative household, heterogeneous entrepreneurs, domestic banks, goods producers, retailers, and a central bank. The core financial friction resides in the entrepreneurial sector, where credit-constrained firms face idiosyncratic investment efficiency shocks and can hold a recurrent bubble asset as a storage of net worth. Domestic banks intermediate both domestic savings and foreign capital to finance these entrepreneurs, subject to reserve requirements. Nominal rigidities are introduced in both the domestic and export retail markets. A central bank conducts active conventional monetary and may adjust the reserve requirement from a macroprudential perspective. Figure 1 illustrates the model structure.

2.1. Households

In each period, the income of the representative household comprises labor income and dividends from banks, entrepreneurs, capital goods producers, and retailers. The household consumes domestic final goods and saves through bank deposits and purchases of stock shares issued by

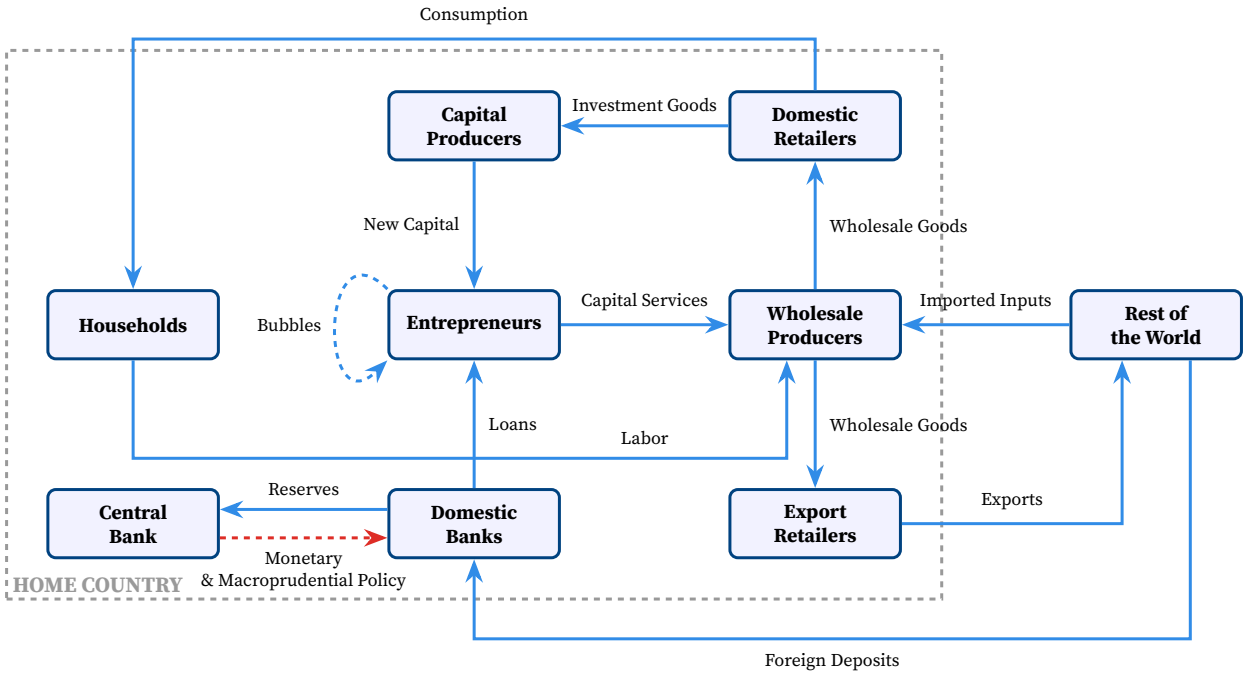


FIGURE 1. Structure of the Small Open Economy

entrepreneurs. The household maximizes its expected lifetime utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\ln(C_t - hC_{t-1}) - \kappa \frac{N_t^{1+\varphi}}{1+\varphi} \right]$$

subject to the budget constraint

$$P_t C_t + B_{ht} + \int \iota_{jt}(V_{jt} - D_{jt})\, dj = W_t N_t + R_{t-1} B_{h,t-1} + \int \iota_{j,t-1} V_{jt}\, dj + D_t^b + D_t^k + D_t^r + T_t, \tag{1}$$

where C_t is consumption, N_t is labor, and β is the subjective discount factor. h captures the strength of consumption habits, $1/\varphi$ is the Frisch elasticity of labor supply, and κ is the weight on labor disutility. D_t^b, D_t^k, and D_t^r represent the profits of domestic banks, capital producers, and retailers operating in both domestic and export markets. T_t is the government transfer. V_{jt} and D_{jt} denote the stock value and dividend issuance of the firm managed by entrepreneur j. $B_{h,t-1}$ and $\iota_{j,t-1}$ denote the deposits and the shares of firm j held by the household at the beginning of period t. R_t denotes the deposit rate between periods t and $t+1$, P_t is the nominal price of consumption goods, and W_t is the nominal wage.

Let Λ_t be the Lagrange multiplier associated with the budget constraint. The marginal utility of consumption is

$$\Lambda_t = \frac{1}{P_t} \left(\frac{1}{C_t - hC_{t-1}} - \mathbb{E}_t \beta \frac{h}{C_{t+1} - hC_t} \right), \tag{2}$$

and the labor supply is determined by

$$\Lambda_t W_t = \kappa N_t^{\varphi}. \tag{3}$$

Suppose that the household cannot borrow from banks so that $B_{ht} \geq 0$. The first-order condition with respect to deposits implies

$$\frac{1}{R_t} \geq \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t}. \tag{4}$$

2.2. Banks

Banks act as intermediaries for both domestic and international financial transactions. In each period t, the representative bank accepts deposits B_t from domestic savers and B_t^* from foreign investors, and allocates these funds to loans L_t for domestic entrepreneurs and reserves RE_t held at the central bank. For simplicity, we assume that banks have zero net worth. The balance sheet constraint of the representative bank is given by

$$L_t + RE_t = B_t + e_t B_t^*, \tag{5}$$

where e_t is the nominal exchange rate defined as the amount of domestic currency in exchange for one unit of foreign currency. The corresponding flow-of-funds constraint is

$$D_{t+1}^b = R_{lt} L_t + RE_t - R_t B_t - e_{t+1} R_t^* B_t^*, \tag{6}$$

where R_{lt} denotes the loan rate, and R_t^* is the international borrowing rate faced by domestic banks. Reserves at the central bank earn no interest and effectively serve as high-powered money or the monetary base.

To close the small open economy model, we follow Schmitt-Grohé and Uribe (2003) and assume that the international borrowing rate, R_t^*, equals the foreign interest rate r_t^* plus a debt-elastic country-risk premium:

$$R_t^* = r_t^* + g\left(\frac{e_t B_t^*}{P_{zt} Z_t}\right). \tag{7}$$

The premium component is defined as

$$g\left(\frac{e_t B_t^*}{P_{zt} Z_t}\right) \equiv \Omega\left[\exp\left(\frac{e_t B_t^*}{P_{zt} Z_t} - \underline{B}^*\right) - 1\right],$$

where $\Omega > 0$ and $\underline{B}^* \equiv \frac{e B^*}{P_z Z}$, that is, the country-risk premium is assumed to be an increasing and convex function of the deviation of the external debt-to-output ratio from its steady-state target. A larger gap results in a higher marginal cost of accepting deposits from foreign investors. The

foreign interest rate r_t^* is exogenous and follows

$$\ln r_t^* = (1-\rho_{r^*})\ln r^* + \rho_{r^*}\ln r_{t-1}^* + \varepsilon_{r^*t}, \tag{8}$$

where $\rho_{r^*} \in (0,1)$ is the persistence parameter, r^* is its steady-state value, and ε_{r^*t} is a white noise process with standard deviation σ_{r^*}.

The representative bank aims to maximize the expected discounted value of its dividend payouts

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\Lambda_t}{\Lambda_0} D_t^b$$

subject to (5), (6) and (7). In addition, assume that the bank faces a reserve requirement

$$RE_t \geq rr_t(B_t + e_t B_t^*),$$

where rr_t is the required reserve ratio that can be adjusted by the central bank counter-cyclically for macroprudential purposes.

Clearly, as long as $R_{lt} > R_t$, the reserve requirement binds so that

$$RE_t = rr_t(B_t + e_t B_t^*), \tag{9}$$

which also gives rise to the relationship between R_{lt} and R_t as

$$R_{lt} = \frac{R_t - rr_t}{1 - rr_t}. \tag{10}$$

This relationship implies that the deposit rate must satisfy $R_t > 1$ in order for $R_{lt} > R_t$ to hold in equilibrium.

The first-order condition with respect to B_t^* yields the familiar uncovered interest rate parity (UIP) condition:

$$\mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t}\left[R_t - \frac{e_{t+1}}{e_t}R_t^*\right] = \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t}\left[\frac{e_{t+1}}{e_t}\Omega \exp\left(\frac{e_t B_t^*}{P_{zt}Z_t} - \underline{B}^*\right)\frac{e_t B_t^*}{P_{zt}Z_t}\right]. \tag{11}$$

2.3. Entrepreneurs

There is a continuum of entrepreneurs indexed by $j \in [0,1]$, each managing a firm that accumulates physical capital K_{jt}. These firms provide capital services to wholesale goods producers and earn capital rent R_{kt} in return. Assume that the law of motion for firm j's capital is

$$K_{jt} = (1-\delta)K_{j,t-1} + \varepsilon_{jt} I_{jt}, \tag{12}$$

where $\delta \in (0,1)$ is the depreciation rate of capital after it is used by the wholesale goods producers, I_{jt} is the investment of firm j in period t, and ε_{jt} represents a firm-specific investment efficiency shock.

This shock is independent and identically distributed across firms and over time. It has a cumulative distribution function $F(\cdot)$ and a probability density function $f(\cdot)$ with support $[\varepsilon_{\min}, \varepsilon_{\max}] \subset [0, \infty)$. Investment is assumed to be irreversible such that $I_{jt} \geq 0$.

The entrepreneur j chooses to save $B_{jt} \geq 0$ in banks at the deposit rate R_t and borrow $L_{jt} \geq 0$ at the lending rate R_{lt}. Each entrepreneur is endowed with $\delta_h \in (0,1)$ units of an intrinsically useless bubble asset at the beginning of each period t. Its nominal price is denoted by $P^h_{t|t}$. Free disposal ensures this price is nonnegative. Assume that in each period, a fraction δ_h of each vintage of the bubble asset loses its value due to depreciation. This implies that the total outstanding amount of bubble assets remains constant and is normalized to one. Our modeling of such recurrent bubbles follows Dong, Miao, and Wang (2020) and Dong, Mei, and Xiao (2025), which makes possible the introduction of a non-fundamental sentiment shock that drives movements in asset bubbles.

Entrepreneurs can trade each vintage of the bubble asset. Let $H_{j,t|t-k}$ represent the period t holding choice of the bubble asset endowed in period $t-k$. By convention, we set $H_{j,t|t} = \delta_h/(1-\delta_h)$. Let $P^h_{t|t-k}$ denote the period t nominal price of the bubble asset of vintage $t-k$. Assume that entrepreneurs cannot short the bubble asset such that

$$H_{j,t|t-k} \geq 0, \quad \text{for } k = 0, 1, 2, \ldots$$

Assume that entrepreneurs face borrowing constraints due to imperfect contract enforcement. Following Dong, Jiao, and Wang (2025), the constraint takes the form

$$L_{jt} \leq \mu P_{kt} K_{j,t-1}, \tag{13}$$

where $\mu \in (0,1)$ is a parameter that represents the extent to which capital can back firms' borrowing, and P_{kt} denotes the nominal price of the capital goods.

The flow-of-funds constraint of entrepreneur j can be written as

$$\begin{aligned} D_{jt} = {} & R_{kt} K_{j,t-1} - P_{kt} I_{jt} + R_{t-1} B_{j,t-1} - B_{jt} \\ & + L_{jt} - R_{l,t-1} L_{j,t-1} + \sum_{k=0}^{\infty} P^h_{t|t-k} \left[(1-\delta_h) H_{j,t-1|t-k} - H_{j,t|t-k} \right], \end{aligned} \tag{14}$$

where D_{jt} denotes the dividend issuance. Suppose equity finance is prohibitively costly and the firm cannot issue any new equity so that[1]

$$D_{jt} \geq 0.$$

Let $V_{jt}(\cdot)$ denote the value function for firm j in period t. We can express the entrepreneur's

[1]The key insights of this paper will not change qualitatively as long as new equity issues are sufficiently constrained. See Miao, Wang, and Xu (2015) and Miao and Wang (2018) for related discussions.

dynamic optimization problem by the Bellman equation

$$
\begin{aligned}
&V_{jt}\left(\varepsilon_{jt}, K_{j,t-1}, B_{j,t-1}, L_{j,t-1}, \{H_{j,t-1|t-k}\}_{k=0}^{\infty}\right) \\
&\quad = \max_{\{I_{jt}, B_{jt}, L_{jt}, H_{j,t|t-k}, D_{jt}\geq 0\}} \left\{ D_{jt} + E_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} V_{j,t+1}\left(\varepsilon_{j,t+1}, K_{jt}, B_{jt}, L_{jt}, \{H_{j,t|t-k}\}_{k=0}^{\infty}\right)\right\},
\end{aligned} \tag{15}
$$

subject to the constraints (12), (13) and (14).

Define the Tobin's marginal Q as

$$
Q_t \equiv \frac{\partial}{\partial K_{jt}} \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} V_{j,t+1}\left(\varepsilon_{j,t+1}, K_{jt}, B_{jt}, L_{jt}, \{H_{j,t|t-k}\}_{k=0}^{\infty}\right).
$$

The following proposition characterizes the entrepreneur j's optimal decisions, with the proof left to Appendix A.1.

PROPOSITION 1. *In an equilibrium with $R_{lt} > R_t > 1$, there exist two thresholds $\bar{\varepsilon}_t$ and $\bar{\varepsilon}_{lt}$ of investment efficiency such that the decision rules of an entrepreneur j are given by $I_{jt} = 0$ if $\varepsilon_{jt} \leq \bar{\varepsilon}_t$ and*

$$
\begin{aligned}
I_{jt} = \frac{1}{P_{kt}} \Bigg[& R_{kt} K_{j,t-1} + R_{t-1} B_{j,t-1} - B_{jt} + L_{jt} - R_{l,t-1} L_{j,t-1} \\
& + \sum_{k=0}^{\infty} P_{t|t-k}^{h} \left((1-\delta_h) H_{j,t-1|t-k} - H_{j,t|t-k} \right) \Bigg], \quad \text{if } \varepsilon_{jt} > \bar{\varepsilon}_t,
\end{aligned}
$$

$$
B_{jt} = \begin{cases} \text{indeterminate}, & \text{if } \varepsilon_{jt} \leq \bar{\varepsilon}_t \\ 0, & \text{if } \varepsilon_{jt} > \bar{\varepsilon}_t \end{cases},
$$

$$
L_{jt} = \begin{cases} 0, & \text{if } \varepsilon_{jt} \leq \bar{\varepsilon}_{lt} \\ \mu P_{kt} K_{j,t-1}, & \text{if } \varepsilon_{jt} > \bar{\varepsilon}_{lt} \end{cases},
$$

$$
H_{j,t|t-k} = \begin{cases} \text{indeterminate}, & \text{if } \varepsilon_{jt} \leq \bar{\varepsilon}_t \\ 0, & \text{if } \varepsilon_{jt} > \bar{\varepsilon}_t \end{cases}, \quad \text{for } k = 0, 1, 2, \ldots
$$

where the thresholds are given by

$$
\bar{\varepsilon}_t = \frac{P_{kt}}{Q_t}, \tag{16}
$$

$$
\bar{\varepsilon}_{lt} = \frac{R_{lt}}{R_t} \bar{\varepsilon}_t, \tag{17}
$$

and Q_t, R_t and $P^h_{t|t-k}$ satisfy

$$Q_t = \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} \Bigg[R_{k,t+1} + (1-\delta) Q_{t+1} + R_{k,t+1} \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) + \mu P_{k,t+1} \int_{\bar{\varepsilon}_{l,t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - \frac{R_{l,t+1}}{R_{t+1}} \right) dF(\varepsilon) \Bigg], \tag{18}$$

$$\frac{1}{R_t} = \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} \left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) \right], \tag{19}$$

$$P^h_{t|t-k} = (1-\delta_h) \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} P^h_{t+1|t-k} \left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) \right]. \tag{20}$$

Proposition 1 demonstrates the existence of two investment efficiency thresholds, $\bar{\varepsilon}_t$ and $\bar{\varepsilon}_{lt}$, which govern the entrepreneur's optimal behavior. A firm undertakes investment if and only if its idiosyncratic efficiency exceeds the first threshold, $\varepsilon_{jt} > \bar{\varepsilon}_t = \frac{P_{kt}}{Q_t}$. This condition is a version of Tobin's rule: since one unit of investment costs P_{kt} and generates ε_{jt} units of capital valued at Q_t, the firm invests when the marginal benefit, $Q_t \varepsilon_{jt}$, exceeds the marginal cost. To finance this investment, the firm first exhausts its internal funds by liquidating all deposits and bubble assets. Because the lending rate R_{lt} exceeds the deposit rate R_t, the firm only resorts to external bank credit when investment is sufficiently profitable—specifically, when its efficiency exceeds the higher threshold $\bar{\varepsilon}_{lt} = \frac{R_{lt}}{R_t} \bar{\varepsilon}_t$. The wedge between these two thresholds represents the external finance premium, analogous to that in Bernanke, Gertler, and Gilchrist (1999). Once the borrowing threshold is crossed, the firm borrows up to its maximum credit limit.

Equations (18), (19), and (20) characterize the pricing of capital, deposits, and the bubble asset, respectively. Beyond standard fundamental components, these equations include integral terms representing the liquidity premia generated by financial frictions. In the capital pricing equation (18), two distinct liquidity premia emerge. The first integral, which also appears in equations (19) and (20), captures the premium associated with internal funds reallocated to efficient investment whenever $\varepsilon_{jt} > \bar{\varepsilon}_{t+1}$. The second integral represents an additional collateral premium: the marginal benefit of capital in relaxing the credit constraint for firms with investment efficiency $\varepsilon_{jt} > \bar{\varepsilon}_{l,t+1}$ to warrant external finance. Finally, the factor $(1-\delta_h)$ in the bubble pricing equation (20) accounts for the physical depreciation of the bubble's value over time.

2.4. Wholesale Goods Producers

Competitive wholesale goods producers operate a constant returns to scale technology. They rent physical capital from entrepreneurs and combine it with labor and imported foreign materials to

produce wholesale goods. The production function is given by:

$$Z_t = K_{t-1}^{\alpha}(A_t N_t)^{1-\alpha-\gamma} M_t^{\gamma}, \quad \alpha, \gamma \in (0,1), \quad \alpha + \gamma \in (0,1), \tag{21}$$

where K_{t-1} is the capital rented from entrepreneurs, N_t is the labor input, and M_t is the imported inputs. The term A_t represents the labor-augmented aggregate productivity level. Following Miao, Wang, and Zhou (2022) and Dong, Jiao, and Wang (2025), assume that A_t is composed of a trend component A_t^g and a transitory shock a_t:

$$A_t = A_t^g \exp(a_t),$$

where the trend productivity follows $A_t^g = A_{t-1}^g \exp(g)$, and the short-run productivity follows

$$a_t = \rho_a a_{t-1} + \varepsilon_{at}, \tag{22}$$

with ε_{at} being a white noise process with standard deviation σ_a.

Wholesale firms solve a static profit maximization problem each period. They choose demand for capital K_{t-1}, labor N_t, and foreign inputs M_t to maximize their profit given by

$$P_{zt} Z_t - R_{kt} K_{t-1} - W_t N_t - e_t P_t^* M_t,$$

where P_{zt} is the price of wholesale goods, and P_t^* is the exogenous price of imported inputs in foreign currency, assumed to be equal to the foreign CPI. First-order conditions with respect to capital, labor, and imported inputs imply that factor prices equal their marginal revenue products:

$$R_{kt} = \alpha \frac{P_{zt} Z_t}{K_{t-1}}, \tag{23}$$

$$W_t = (1-\alpha-\gamma) \frac{P_{zt} Z_t}{N_t}, \tag{24}$$

$$e_t P_t^* = \gamma \frac{P_{zt} Z_t}{M_t}. \tag{25}$$

As is standard in competitive market with constant returns to scale technology, wholesale goods producers earn zero profit in equilibrium.

2.5. Capital Producers

A representative capital goods producer utilizes final goods as inputs to produce new physical capital, which it sells to entrepreneurs at price P_{kt}, subject to quadratic investment adjustment costs. The objective of the capital producer is to maximize the expected present discounted value of dividends:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\Lambda_t}{\Lambda_0} D_t^k,$$

where D_t^k is given by

$$D_t^k = P_{kt} I_t - \left[1 + \frac{\Omega_k}{2}\left(\frac{I_t}{I_{t-1}} - \exp(g)\right)^2\right] P_t I_t, \tag{26}$$

and $\Omega_k > 0$ governs the magnitude of the adjustment costs. The first-order condition with respect to I_t is given by:

$$\begin{aligned} P_{kt} = &\left[1 + \frac{\Omega_k}{2}\left(\frac{I_t}{I_{t-1}} - \exp(g)\right)^2\right] P_t + \Omega_k P_t \left(\frac{I_t}{I_{t-1}} - \exp(g)\right)\frac{I_t}{I_{t-1}} \\ &- \Omega_k \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} P_{t+1}\left(\frac{I_{t+1}}{I_t} - \exp(g)\right)\left(\frac{I_{t+1}}{I_t}\right)^2. \end{aligned} \tag{27}$$

2.6. Domestic Retailers

There exists a continuum of monopolistically competitive retailers in the domestic market, indexed by $r \in [0,1]$. In each period t, they purchase wholesale goods at the nominal price P_{zt}, differentiate them at no cost, and sell the differentiated good r to domestic agents at the nominal price P_{rt}.

Intermediate goods are transformed into final goods according to a standard CES aggregator:

$$Y_t = \left[\int_0^1 Y_{rt}^{\frac{\psi-1}{\psi}} dr\right]^{\frac{\psi}{\psi-1}}, \quad \psi > 1.$$

Cost minimization yields the downward-sloping demand curve for each retailer r:

$$Y_{rt} = \left(\frac{P_{rt}}{P_t}\right)^{-\psi} Y_t, \tag{28}$$

where the aggregate price index (domestic CPI) is given by:

$$P_t \equiv \left[\int_0^1 P_{rt}^{1-\psi} dr\right]^{\frac{1}{1-\psi}}.$$

To introduce nominal rigidities, we follow the staggered pricing mechanism of Calvo (1983), augmented by indexation to lagged inflation as in Christiano, Eichenbaum, and Evans (2005). Specifically, in any given period, a retailer can re-optimize its price only with probability $1 - \chi$. If a firm cannot re-optimize, its price is automatically indexed to the previous period's domestic CPI inflation rate. A re-optimizing retailer chooses its reset price, $\tilde{P}_{rt}$, to maximize the expected present discounted value of future profits under this setup:

$$\max_{\tilde{P}_{rt}} \mathbb{E}_t \sum_{i=0}^{\infty} \chi^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t}\left(\tilde{P}_{rt} \prod_{k=1}^{i} \pi_{t+k-1} - P_{z,t+i}\right) Y_{r,t+i},$$

where $\pi_t \equiv \frac{P_t}{P_{t-1}}$ is the period t gross domestic CPI inflation rate, subject to the sequence of demand constraints, which now reflect the indexed price:

$$Y_{r,t+i} = \left(\frac{\tilde{P}_{rt} \prod_{k=1}^{i} \pi_{t+k-1}}{P_{t+i}} \right)^{-\psi} Y_{t+i}.$$

The first-order condition with respect to $\tilde{P}_{rt}$ yields the optimal pricing rule:

$$\tilde{P}_{rt} = \frac{\psi}{\psi - 1} \frac{\mathbb{E}_t \sum_{i=0}^{\infty} \chi^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t} Y_{t+i} P_{z,t+i} P_{t+i}^{\psi} \left(\prod_{k=1}^{i} \pi_{t+k-1} \right)^{-\psi}}{\mathbb{E}_t \sum_{i=0}^{\infty} \chi^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t} Y_{t+i} P_{t+i}^{\psi} \left(\prod_{k=1}^{i} \pi_{t+k-1} \right)^{1-\psi}},$$

which can be rewritten in a recursive form:

$$\tilde{P}_{rt} = \frac{\psi}{\psi - 1} \frac{\Gamma_{nt}}{\Gamma_{dt}}, \tag{29}$$

where the auxiliary variables Γ_{nt} and Γ_{dt} evolve according to:

$$\Gamma_{nt} = \Lambda_t P_{zt} P_t^{\psi} Y_t + \chi \beta \pi_t^{-\psi} \mathbb{E}_t \Gamma_{n,t+1}, \tag{30}$$

$$\Gamma_{dt} = \Lambda_t P_t^{\psi} Y_t + \chi \beta \pi_t^{1-\psi} \mathbb{E}_t \Gamma_{d,t+1}. \tag{31}$$

The aggregate price level evolves according to a weighted average of the newly optimized prices and the indexed prices of the non-optimizing firms from the previous period:

$$P_t = \left[(1-\chi) \tilde{P}_{rt}^{1-\psi} + \chi (\pi_{t-1} P_{t-1})^{1-\psi} \right]^{\frac{1}{1-\psi}}. \tag{32}$$

2.7. Export Retailers

Similar to those in the domestic market, there exists a continuum of export retailers, indexed by $r^* \in [0,1]$. These retailers purchase wholesale goods in the domestic market at the nominal price P_{zt}, differentiate them at no cost, and sell them in the foreign market at the nominal price P_{rt}^* in foreign currency.

Each export retailer r^* faces the following downward-sloping foreign demand curve:

$$Y_{rt}^* = \left(\frac{P_{rt}^*}{P_t^*} \right)^{-\psi^*} Y_t^*, \quad \psi^* > 1 \tag{33}$$

where ψ^* is the price elasticity, and Y_t^* represents the exogenous aggregate foreign demand. To ensure a balanced growth path, assume that Y_t^* follows

$$Y_t^* = A_t^g y^*$$

where y^* represents the size of the foreign economy.

Following Calvo (1983), we assume that each export retailer can change its price optimally in period t with probability $1-\chi^*$. Then, a retailer chooses $\tilde{P}^*_{rt}$ optimally to solve the problem:

$$\max_{\tilde{P}^*_{rt}} \mathbb{E}_t \sum_{i=0}^{\infty} {\chi^*}^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t} \left(e_{t+i} \tilde{P}^*_{rt} - P_{z,t+i} \right) Y^*_{r,t+i},$$

subject to the sequence of demand constraints (33).

The first-order condition with respect to $\tilde{P}^*_{rt}$ yields the optimal pricing rule:

$$\tilde{P}^*_{rt} = \frac{\psi^*}{\psi^* - 1} \frac{\mathbb{E}_t \sum_{i=0}^{\infty} {\chi^*}^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t} Y^*_{t+i} P_{z,t+i} {P^*_{t+i}}^{\psi^*}}{\mathbb{E}_t \sum_{i=0}^{\infty} {\chi^*}^i \beta^i \frac{\Lambda_{t+i}}{\Lambda_t} Y^*_{t+i} e_{t+i} {P^*_{t+i}}^{\psi^*}},$$

which can be rewritten in a recursive form:

$$\tilde{P}^*_{rt} = \frac{\psi^*}{\psi^* - 1} \frac{\Gamma^*_{nt}}{\Gamma^*_{dt}}, \tag{34}$$

where the auxiliary variables Γ^*_{nt} and Γ^*_{dt} evolve according to:

$$\Gamma^*_{nt} = \Lambda_t P_{zt} {P^*_t}^{\psi^*} Y^*_t + \chi^* \beta \mathbb{E}_t \Gamma^*_{n,t+1}, \tag{35}$$

$$\Gamma^*_{dt} = \Lambda_t e_t {P^*_t}^{\psi^*} Y^*_t + \chi^* \beta \mathbb{E}_t \Gamma^*_{d,t+1}. \tag{36}$$

2.8. Central Bank

We assume that the central bank acts as both the monetary and regulatory authority, conducting monetary policy and macroprudential regulation through reserve requirements.

For monetary policy, we consider a Taylor-type interest rate rule where the central bank sets the nominal interest rate R_t in response to current domestic inflation and output fluctuations, potentially augmented to respond to asset price fluctuations. The monetary policy rule is given by:

$$\ln\left(\frac{R_t}{R}\right) = \phi_\pi \ln\left(\frac{\pi_t}{\pi}\right) + \phi_z \ln\left(\frac{Z_t}{Z}\right) + \phi_f \ln\left(\frac{P_{st}/P_t}{P_s/P}\right), \tag{37}$$

where R is the steady-state nominal interest rate, π is the long-run domestic inflation target, Z is the steady-state output level of wholesale goods, P_{st} denotes the aggregate nominal stock price index that contains a bubble component (to be formally defined in the next section), so that P_{st}/P_t represents the real stock price and P_s/P is its steady state level. The parameter ϕ_f governs the intensity of the central bank's "leaning-against-the-wind" monetary policy.

The interest rate rule implies that the money supply is endogenous. Let reserves follow the

dynamics:

$$RE_t = RE_{t-1} \exp(g_{RE,t}), \tag{38}$$

where $g_{RE,t}$ is the endogenous exponential growth rate of the monetary base. We assume that the seigniorage revenue T_t from the newly created money are transferred to households in a lump-sum way.

In addition to the standard interest rate rule, the central bank may actively utilize a reserve requirement policy to achieve its macroprudential objectives. We assume the reserve requirement ratio, rr_t, responds directly to the stock price fluctuations:

$$rr_t - rr = \phi_{rr} \ln\left(\frac{P_{st}/P_t}{P_s/P}\right), \tag{39}$$

where rr is the steady-state reserve requirement ratio, and ϕ_{rr} governs the responsiveness of the macroprudential tool to asset price fluctuations.

Alternative Monetary and Macroprudential Policy Regimes. To examine the effectiveness of different policy instruments in stabilizing the financial market, we consider three alternative policy regimes, distinguished by whether monetary or macroprudential policy actively responds to asset price fluctuations:

a. **Taylor Rule (TR):** Monetary policy responds only to inflation and the output fluctuations ($\phi_\pi > 1, \phi_z > 0, \phi_f = 0$), and the reserve requirement is fixed at its steady state ($\phi_{rr} = 0$).
b. **Augmented Taylor Rule (ATR):** Monetary policy actively responds to asset price fluctuations in addition to inflation and the output ($\phi_\pi > 1, \phi_z > 0, \phi_f > 0$), while macroprudential policy remains inactive ($\phi_{rr} = 0$).
c. **Macroprudential Policy (MACRO):** The reserve requirement is used to control credit supply in response to asset price fluctuations ($\phi_{rr} > 0$), while monetary policy is strictly devoted to macroeconomic stabilization via the Taylor rule ($\phi_\pi > 1, \phi_z > 0, \phi_f = 0$).

2.9. Aggregation and Equilibrium

The market-clearing condition for bank deposits is

$$\int B_{jt}\, dj + B_{ht} = B_t.$$

From equations (4) and (19), it is clear that $B_{ht} = 0$ in equilibrium because the deposit rate is too low[2]. Assuming $B_{h,-1} = 0$, it follows that $B_{ht} = 0$ for all t so that $\int B_{jt}\, dj = B_t$.

[2]Similarly, as implied by equation (20), households will not hold any bubble asset even though they are allowed to do so. It is a common feature that households are crowded out of the bond market in models where such an asset offers a liquidity premium to other market participants, such as bankers or entrepreneurs. For models with similar crowding-out mechanisms but without asset bubbles, see Kiyotaki and Moore (2005) for an RBC environment and

The market-clearing condition for the bubble asset of vintage k is

$$\int H_{j,t|t-k}\, dj = \delta_h (1-\delta_h)^k, \quad k = 0, 1, 2, \ldots,$$

under which we define an index for aggregate pre-existing bubbles P^h_{et} as:

$$P^h_{et} = \sum_{k=1}^{\infty} P^h_{t|t-k}(1-\delta_h)^k \delta_h.$$

Denote the total size of newly created bubbles by $P^h_{nt} = \delta_h P^h_{t|t}$ and define the economy's aggregate bubble index as the sum of pre-existing and new bubbles, $P^h_t = P^h_{et} + P^h_{nt}$. Using these definitions, we can aggregate equation (20) across vintages and obtain the asset pricing equation for the aggregate bubbles as:

$$P^h_t = \mathbb{E}_t \beta \frac{\Lambda_{t+1}}{\Lambda_t} P^h_{e,t+1} \left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) \right]. \tag{40}$$

Note that the expected pre-existing bubble tomorrow, $P^h_{e,t+1}$, appears on the right-hand side of equation (40) because all bubbles that make up the aggregate index today will become pre-existing bubbles tomorrow.

In equilibrium, the sizes of pre-existing bubbles and new bubbles cannot be independently determined, which implies the existence of a continuum of bubbly equilibria. Here we follow Dong, Miao, and Wang (2020) and assume that pre-existing bubbles account for a stochastic fraction of the aggregate bubble index, such that

$$P^h_{et} = s_t P^h_t, \tag{41}$$

where the fraction $s_t \in (0,1)$ is interpreted as an exogenous sentiment shock that drives the fluctuations of aggregate bubbles. Assume that s_t evolves according to:

$$\ln s_t = (1-\rho_s) \ln s + \rho_s \ln s_{t-1} + \varepsilon_{st}, \tag{42}$$

where $\rho_s \in (0,1)$ governs the persistence of the sentiment bubble shock, $s \in (0,1)$ is the steady-state value of s_t, and ε_{st} is a white noise process with standard deviation σ_s.

The market-clearing conditions for bank loans and physical capital are:

$$\int L_{jt}\, dj = L_t, \quad \int K_{jt}\, dj = K_t.$$

By Proposition 1, highly efficient firms (those with $\varepsilon_{jt} \geq \bar{\varepsilon}_{lt}$) are not fully self-financed and will optimally borrow up to the credit limit in (13). Aggregating these borrowing constraints across all

Vlieghe (2010) for a New Keynesian setting.

lending banks yields:

$$L_t = \mu P_{kt} K_{t-1}\big(1 - F(\bar{\varepsilon}_{lt})\big). \tag{43}$$

Combining the bank balance sheet (5) with the reserve requirement (9), equation (43) implies the following aggregate demand for reserves:

$$RE_t = \frac{rr_t}{1 - rr_t} \mu P_{kt} K_{t-1} \left[1 - F(\bar{\varepsilon}_{lt})\right], \tag{44}$$

which serves as the equilibrium monetary demand condition.

Define aggregate entrepreneurial investment as $I_t \equiv \int I_{jt}\, dj$. Applying the optimal decision rules from Proposition 1, we can express aggregate investment as:

$$I_t = \frac{1}{P_{kt}} \left[NW_t\big(1 - F(\bar{\varepsilon}_t)\big) + L_t\right], \tag{45}$$

where

$$NW_t \equiv R_{kt} K_{t-1} + R_{t-1} B_{t-1} - R_{l,t-1} L_{t-1} + P_t^h \tag{46}$$

denotes the aggregate net worth of the entrepreneurial sector. Equation (45) highlights the two sources of investment financing: NW_t represents the total *internal* funds available to entrepreneurs, and L_t represents the *external* credit. Furthermore, equations (43) and (45) decomposes aggregate investment into an *intensive* and an *extensive margin*. The magnitudes of aggregate net worth, NW_t, and the aggregate collateralizable value of physical capital, $\mu P_{kt} K_{t-1}$, determine the *intensive margin*, while the idiosyncratic efficiency thresholds, $\bar{\varepsilon}_t$ and $\bar{\varepsilon}_{lt}$, govern the *extensive margin* by determining the mass of entrepreneurs who actively invest and borrow.

Substituting out the financial position terms using equations (5), (9), and (10), we can rewrite equation (46) as:

$$NW_t = R_{kt} K_{t-1} + RE_{t-1} - R_{t-1} e_{t-1} B^*_{t-1} + P_t^h. \tag{47}$$

Equation (47) illustrates that aggregate entrepreneurial net worth comprises the fundamental rental income on physical capital, the reserve holdings, and the value of the aggregate bubble, net of the cost of servicing foreign debt.

Aggregating equation (12) across all entrepreneurs yields the law of motion for aggregate physical capital:

$$K_t = (1 - \delta) K_{t-1} + \left(I_t - \frac{L_t}{P_{kt}}\right) \frac{\int_{\bar{\varepsilon}_t}^{\varepsilon_{\max}} \varepsilon\, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_t)} + \frac{L_t}{P_{kt}} \frac{\int_{\bar{\varepsilon}_{lt}}^{\varepsilon_{\max}} \varepsilon\, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_{lt})}. \tag{48}$$

Define the aggregate nominal stock price index, P_{st}, as the sum of the stock values of all

individual firms. Aggregating the value function V_{jt}[3] across all entrepreneurs yields:

$$P_{st} \equiv \int_0^1 V_{jt}\, dj = Q_t K_t + B_t - \frac{R_{lt}}{R_t} L_t + P_t^h,$$

which, using equations (5), (9), and (10), can again be rewritten as:

$$P_{st} = Q_t K_t + \frac{1}{R_t} R E_t - e_t B_t^* + P_t^h. \tag{49}$$

Equation (49) indicates that aggregate stock market capitalization equals the fundamental value of physical capital, the present discounted value of central bank reserves, and the aggregate bubble, net of the foreign debt burden. It is also useful to note the structural parallel between P_{st} and aggregate net worth NW_t in equation (47).

The market-clearing condition for wholesale goods requires that total wholesale production equals the combined demand from domestic and export retailers. Integrating the individual retail goods demand curve (28) over all domestic retailers, and similarly the demand curve (33) for export retailers, yields:

$$Z_t = \Delta_{yt} Y_t + X_t, \tag{50}$$

where

$$X_t \equiv \Delta_{yt}^* Y_t^* \tag{51}$$

denotes the aggregate foreign demand for domestic wholesale goods, $\Delta_{yt} \equiv \int_0^1 \left(\frac{P_{rt}}{P_t}\right)^{-\psi} dr$ and $\Delta_{yt}^* \equiv \int_0^1 \left(\frac{P_{rt}^*}{P_t^*}\right)^{-\psi^*} dr^*$ measure the equilibrium price dispersion arising from nominal rigidities in the domestic and foreign markets, respectively. Under the Calvo (1983) staggered pricing mechanism, these two price dispersion indices evolve according to the following recursive laws of motion:

$$\Delta_{yt} = (1-\chi)\left(\frac{\tilde{P}_{rt}}{P_t}\right)^{-\psi} + \chi\left(\frac{\pi_t}{\pi_{t-1}}\right)^{\psi} \Delta_{y,t-1}, \tag{52}$$

$$\Delta_{yt}^* = (1-\chi^*)\left(\frac{\tilde{P}_{rt}^*}{P_t^*}\right)^{-\psi^*} + \chi^* \pi_t^{*\psi^*} \Delta_{y,t-1}^*, \tag{53}$$

where $\pi_t^* \equiv \frac{P_t^*}{P_{t-1}^*}$ denotes the gross inflation rate in the foreign market.

The aggregate resource constraint for domestic final goods requires that total final output be allocated between household consumption and capital investment (inclusive of quadratic adjustment

[3] See the proof of Proposition 1 at Appendix A.1 for the functional form of V_{jt}.

costs):

$$Y_t = C_t + \left[1 + \frac{\Omega_k}{2}\left(\frac{I_t}{I_{t-1}} - \exp(g)\right)^2\right] I_t. \tag{54}$$

Finally, we impose the balance of payments (BOP) equation, which dictates that net exports must equal the net change in the foreign debt position:

$$e_t \Delta^*_{pt} Y^*_t - e_t P^*_t M_t = e_t R^*_{t-1} B^*_{t-1} - e_t B^*_t, \tag{55}$$

where $\Delta^*_{pt} \equiv \int_0^1 P^*_{rt} \left(\frac{P^*_{rt}}{P^*_t}\right)^{-\psi^*} dr^*$ is the export price dispersion index, the law of motion of which is given by

$$\Delta^*_{pt} = (1-\chi^*)\tilde{P}^*_{rt}\left(\frac{\tilde{P}^*_{rt}}{P^*_t}\right)^{-\psi^*} + \chi^* \pi^{*\psi^*}_t \Delta^*_{p,t-1}. \tag{56}$$

The derivation of the BOP condition is left to Appendix A.2 as the proof for the following proposition that characterizes the competitive equilibrium of the small open economy:

PROPOSITION 2. *Let K_{-1}, B^*_{-1}, and RE_{-1} be the initial aggregate capital stock, historical foreign debt, and historical central bank reserves, respectively. Let e_{-1}, R_{-1}, and R^*_{-1} be the historical exchange rate and interest rates, let π_{-1} be the historical domestic inflation rate, and let $\Delta_{y,-1}$, $\Delta^*_{y,-1}$, and $\Delta^*_{p,-1}$ be the initial price dispersions. Let the exogenous processes for productivity, foreign interest rate, and sentiment bubble shock a_t, r^*_t, s_t for $t \geq 0$ evolve according to equations (22), (8), and (42). A competitive equilibrium in this small open economy consists of a sequence of 16 endogenous prices $\{P_t, P_{kt}, P^h_t, P^e_t, P_{st}, P_{zt}, \tilde{P}_{rt}, \tilde{P}^*_{rt}, W_t, R_{kt}, R_{lt}, R^*_t, Q_t, e_t, \pi_t, \Lambda_t\}$; 13 aggregate quantities $\{C_t, N_t, Y_t, Z_t, M_t, X_t, K_t, I_t, L_t, B_t, B^*_t, RE_t, NW_t\}$; 7 auxiliary variables $\{\Delta_{yt}, \Delta^*_{yt}, \Delta^*_{pt}, \Gamma_{nt}, \Gamma_{dt}, \Gamma^*_{nt}, \Gamma^*_{dt}\}$; 4 policy variables $\{R_t, rr_t, g_{RE,t}, T_t\}$; and 2 idiosyncratic decision thresholds $\{\bar{\varepsilon}_t, \bar{\varepsilon}_{lt}\}$ for $t \geq 0$, such that:*

a. ***Optimality:*** *Households, banks, entrepreneurs, wholesale goods producers, capital producers, and retailers solve their respective optimization problems. The resulting factor demands and supplies, along with asset prices and decision thresholds, satisfy equations (2), (3), (5), (7), (10), (11), (16), (17), (18), (19), (21), (23), (24), (25), (27), (29), (30), (31), (32), (34), (35), (36), (40), (43), (44), (45), (47), (49), with the bubble relation given by equation (41).*
b. ***Aggregation and Evolution:*** *Aggregate physical capital and price dispersions evolve according to equations (48), (52), (53), and (56), along with the inflation identity $\pi_t = P_t/P_{t-1}$.*
c. ***Government Policy:*** *The central bank sets the nominal interest rate R_t following the policy rule specified in equation (37), with the endogenous reserve growth rate $g_{RE,t}$ satisfying equation (38), and sets the reserve requirement rr_t following equation (39). The seigniorage revenue is transferred to households according to the identity $T_t = RE_t - RE_{t-1}$.*
d. ***Market Clearing and Balance of Payments:*** *All markets for labor, goods, and assets clear, and the balance of payments holds, satisfying equations (50), (51), (54), and (55).*

We focus on a small open economy and consider the rest of the world as exogenous. Following Dong, Jiao, and Wang (2025), to ensure that the dynamic responses to shocks are evaluated around a constant steady state, we take foreign prices as the numeraire and normalize the equilibrium with respect to the foreign currency. Specifically, we normalize domestic nominal prices and nominal financial positions by the nominal exchange rate, e_t, to obtain real values expressed in foreign currency. For a generic nominal variable X_t, its normalized value is denoted by a lowercase letter[4], $x_t \equiv \frac{X_t}{e_t}$. We define $\pi_{et} \equiv \frac{e_t}{e_{t-1}}$ as the depreciation rate of the domestic currency. The full normalized equilibrium system is presented in Appendix B.1.

Moreover, because the economy exhibits a productivity trend, we detrend the system along its balanced growth path by the trend component of labor-augmented TFP, A_t^g. Accordingly, a lowercase letter with a tilde denotes the detrended value of a normalized price, e.g., $\tilde{x}_t \equiv \frac{x_t}{A_t^g}$. The fully detrended system is presented in Appendix B.2.

3. Steady State Analysis

The model admits two types of steady-state equilibria: bubbleless and bubbly, as $P_t^h = 0$ for all t always satisfies equation (40). We drop the time subscript t of a detrended variable to denote its steady-state value. Whenever necessary, we append the subscript f or b to indicate its bubbleless or bubbly steady-state value, respectively.

We start by analyzing several key equilibrium conditions that are common to both the bubbleless and bubbly steady states, before characterizing each of them in turn. Proofs of lemmas and propositions in this section are relegated to Appendix C.

From equation (19), the steady-state deposit rate R can be written as a function of the investment threshold, $\bar{\varepsilon}$:

$$R(\bar{\varepsilon}) = \frac{\pi\beta^{-1}\exp(g)}{1 + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}} - 1\right) dF(\varepsilon)}. \tag{57}$$

It is straightforward to show[5] that $R(\bar{\varepsilon})$ increases with $\bar{\varepsilon}$. To ensure $R > 1$ so that the reserve requirement always binds, we need the following lemma.

LEMMA 1. *Suppose* $\frac{\mathbb{E}[\varepsilon]}{\varepsilon_{\min}} > \pi\beta^{-1}\exp(g) > 1$. *Then, there exists a unique threshold* $\underline{\varepsilon} \in (\varepsilon_{\min}, \varepsilon_{\max})$ *such that*

$$1 + \int_{\underline{\varepsilon}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\underline{\varepsilon}} - 1\right) dF(\varepsilon) = \pi\beta^{-1}\exp(g). \tag{58}$$

Equation (58) defines the lower boundary $\underline{\varepsilon}$ where $R(\underline{\varepsilon}) = 1$, so that $R > 1$ is guaranteed for any $\bar{\varepsilon} > \underline{\varepsilon}$.

[4]The normalized marginal utility of wealth is defined as $\lambda_t \equiv \Lambda_t e_t$.
[5]See the proof of Lemma 1 at Appendix C.1.

Combining equations (10) and (17), we can express the steady-state borrowing threshold, $\bar{\varepsilon}_l$, as a function of $\bar{\varepsilon}$:

$$\bar{\varepsilon}_l = \left[\frac{1 - rr/R(\bar{\varepsilon})}{1 - rr}\right]\bar{\varepsilon} \equiv \varepsilon_l(\bar{\varepsilon}). \tag{59}$$

It is clear that $\bar{\varepsilon}_l$ is increases wuth $\bar{\varepsilon}$. Also, given $\bar{\varepsilon}$, $\bar{\varepsilon}_l$ increases with rr under $R > 1$.

From equation (18), the steady-state real rental rate of capital can also be expressed as a function of $\bar{\varepsilon}$:

$$\frac{r_k}{p_k}(\bar{\varepsilon}) = \frac{\exp(g) - \beta(1-\delta) - \mu\beta\int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}}(\varepsilon - \varepsilon_l(\bar{\varepsilon}))dF(\varepsilon)}{\beta\left[\bar{\varepsilon} + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}}(\varepsilon - \bar{\varepsilon})dF(\varepsilon)\right]}. \tag{60}$$

The following lemma characterizes the behavior of this rental rate.

LEMMA 2. *For a sufficiently small $\mu > 0$, $\frac{r_k}{p_k}(\bar{\varepsilon})$ decreases with $\bar{\varepsilon}$.*

Evaluating the UIP condition (11) at steady-state yields:

$$\frac{b^*}{p_z z} = \frac{1}{\Omega}\left[\frac{R(\bar{\varepsilon})}{\pi} - r^*\right] \equiv \frac{b^*}{p_z z}(\bar{\varepsilon}). \tag{61}$$

This indicates that the steady-state external debt-to-output ratio increases with the real interest rate differential. When the foreign interest rate, r^*, is higher (lower) than the domestic real interest rate, $\frac{R(\bar{\varepsilon})}{\pi}$, the economy experiences a capital account surplus (deficit), corresponding to a net capital outflow (inflow). An increase in the foreign interest rate reduces capital inflows and increasing capital outflows.

3.1. Bubbleless Steady State

We characterize the bubbleless steady state in the following proposition.

PROPOSITION 3. *Suppose that the assumptions in Lemma 1 and Lemma 2 hold. Define the function $\mathcal{H}(\bar{\varepsilon})$ as follows:*

$$\begin{aligned}\mathcal{H}(\bar{\varepsilon}) = &\left\{\frac{r_k}{p_k}(\bar{\varepsilon})\left[1 - \frac{1}{\alpha\exp(g)}\frac{R(\bar{\varepsilon})}{\pi}\frac{b^*}{p_z z}(\bar{\varepsilon})\right]\right.\\ &\left.+\frac{1}{\pi}\frac{rr}{1-rr}\frac{\mu}{\exp(g)}\left(1 - F(\varepsilon_l(\bar{\varepsilon}))\right)\right\}\int_{\bar{\varepsilon}}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon) + \mu\int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon),\end{aligned} \tag{62}$$

where $R(\bar{\varepsilon})$, $\varepsilon_l(\bar{\varepsilon})$, $\frac{r_k}{p_k}(\bar{\varepsilon})$, and $\frac{b^*}{p_z z}(\bar{\varepsilon})$ are given by equations (57), (59), (60), and (61), respectively. If, in addition,

$$\mathcal{H}(\underline{\varepsilon}) > \exp(g) - (1-\delta) \tag{63}$$

holds, with $\underline{\varepsilon}$ defined in equation (58), then there exists a unique bubbleless steady state with $R_f \equiv R(\bar{\varepsilon}_f) > 1$, where $\bar{\varepsilon}_f \in (\underline{\varepsilon}, \varepsilon_{\max})$ is the unique solution to the equation

$$\mathcal{H}(\bar{\varepsilon}) = \exp(g) - (1-\delta). \tag{64}$$

The following condition is necessary to ensure that $x_f > 0$ and $c_f > 0$:

$$\begin{aligned} & 1 - \frac{\psi - 1}{\psi}\left\{\alpha + \frac{1}{\pi}\frac{rr}{1 - rr\exp(g)}\frac{\mu\alpha}{\exp(g)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\left(1 - F(\varepsilon_l(\bar{\varepsilon}_f))\right)\right. \\ & \left. - \frac{R(\bar{\varepsilon}_f)}{\pi}\frac{1}{\exp(g)}\frac{b^*}{p_z z}(\bar{\varepsilon}_f) + \mu\alpha\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\frac{1 - F(\varepsilon_l(\bar{\varepsilon}_f))}{1 - F(\bar{\varepsilon}_f)}\right\}\left(1 - F(\bar{\varepsilon}_f)\right) \\ & > \frac{\psi^* - 1}{\psi^*}\left[\frac{b^*}{p_z z}(\bar{\varepsilon}_f)\left(\frac{r^*}{\exp(g)} - 1\right) + \gamma\right] > 0. \end{aligned} \tag{65}$$

Equation (64) is derived by evaluating the aggregate law of motion for physical capital at the steady state. Its right-hand side represents the ratio of aggregate investment by the entire entrepreneurial sector to the capital stock. In the definition of $\mathcal{H}(\bar{\varepsilon})$, the first term represents the portion of investment financed by the net worth of entrepreneurs, and the second term represents the portion financed by bank loans extended to firms with efficiency levels exceeding the borrowing threshold $\varepsilon_l(\bar{\varepsilon})$.

It is straightforward to verify that $\mathcal{H}(\bar{\varepsilon})$ decreases with $\bar{\varepsilon}$ under Lemma 2. By the Intermediate Value Theorem, this guarantees a unique solution $\bar{\varepsilon}_f \in (\underline{\varepsilon}, \varepsilon_{\max})$ to equation (64) under assumption (63). Once this threshold value is obtained, the steady-state values of all other variables can be uniquely determined as detailed in Appendix C.3.

3.2. Bubbly Steady State

We characterize the bubbly steady state in the following proposition.

PROPOSITION 4. *Suppose that there exists a bubbleless equilibrium with the investment threshold $\bar{\varepsilon}_f$ characterized by Proposition 3. Then, for each value of $s \in (0,1)$, there exists a bubbly steady state with $R_b \equiv R(\bar{\varepsilon}_b) > R_f$ if and only if*

$$s\beta\left[1 + \int_{\bar{\varepsilon}_f}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}_f} - 1\right)dF(\varepsilon)\right] > 1, \tag{66}$$

where $\bar{\varepsilon}_b \in \left(\bar{\varepsilon}_f, \varepsilon_{\max}\right)$ is the unique solution to the equation

$$s\beta\left[1 + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}} - 1\right)dF(\varepsilon)\right] = 1. \tag{67}$$

The following condition ensures $x_b > 0$ and $c_b > 0$:

$$1 - \frac{\psi - 1}{\psi}\Bigg\{\alpha + \frac{1}{\pi}\frac{rr}{1-rr}\frac{\mu\alpha}{\exp(g)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_b)\right]^{-1}\left(1 - F(\varepsilon_l(\bar{\varepsilon}_b))\right)$$

$$- \frac{R(\bar{\varepsilon}_b)}{\pi}\frac{1}{\exp(g)}\frac{b^*}{p_z z}(\bar{\varepsilon}_b) + \mu\alpha\left[\frac{r_k}{p_k}(\bar{\varepsilon}_b)\right]^{-1}\frac{1 - F(\varepsilon_l(\bar{\varepsilon}_b))}{1 - F(\bar{\varepsilon}_b)} + \frac{\tilde{p}_b^h}{p_{zb} z_b}\Bigg\}\left(1 - F(\bar{\varepsilon}_b)\right) \tag{68}$$

$$> \frac{\psi^* - 1}{\psi^*}\left[\frac{b^*}{p_z z}(\bar{\varepsilon}_b)\left(\frac{r^*}{\exp(g)} - 1\right) + \gamma\right] > 0,$$

where $R(\bar{\varepsilon})$, $\varepsilon_l(\bar{\varepsilon})$, $\frac{r_k}{p_k}(\bar{\varepsilon})$, and $\frac{b^*}{p_z z}(\bar{\varepsilon})$ are given by equations (57), (59), (60), and (61), respectively, while $\frac{\tilde{p}_b^h}{p_{zb} z_b}$ is given by

$$\frac{\tilde{p}_b^h}{p_{zb} z_b} = \alpha\frac{[\exp(g) - (1-\delta)] - \mathcal{H}(\bar{\varepsilon}_b)}{\int_{\bar{\varepsilon}_b}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_b)\right]^{-1} \tag{69}$$

for $\mathcal{H}(\bar{\varepsilon})$ defined by equation (62).

The interpretation of the bubble existence condition (66) is as follows: The left-hand side represents the benefit of buying one unit of the bubble asset today, evaluated at the bubbleless steady state, and the right-hand side represents the associated cost. In addition to its resale value, the bubble asset commands a liquidity premium due to credit constraints, which corresponds to the integral term in (66). The liquidity premium emerges because an entrepreneur can liquidate their bubble holdings to finance highly efficient investments if their idiosyncratic investment efficiency exceeds the threshold $\bar{\varepsilon}_f$. When the liquidity premium is large enough so that the benefit exceeds the cost in the bubbleless steady state, entrepreneurs have incentives to trade the bubble asset at a positive price, thereby creating a bubble.

Equation (67) is derived from equation (40), the asset pricing condition for the aggregate bubble. Once $\bar{\varepsilon}_b$ is obtained from this equation, the steady-state values of all other variables can be uniquely determined as detailed in Appendix C.4.

Drivers of Bubble Existence. We now examine how the external financial environment, represented by the world interest rate r^*, and domestic policy stances, namely the inflation target π and the reserve requirement rr, affect the bubble existence condition (66) by altering the bubbleless steady state investment threshold $\bar{\varepsilon}_f$.

From Proposation 3, $\bar{\varepsilon}_f$ is pinned down by the intersection of the downward-sloping $\frac{i}{k}$ schedule, $\mathcal{H}(\bar{\varepsilon})$, and the steady-state capital replacement requirement. Any economic force that erodes entrepreneurial net worth or tightens credit access will shift the $\mathcal{H}(\bar{\varepsilon})$ schedule downward. As a result, to clear the capital market, the equilibrium investment threshold $\bar{\varepsilon}_f$ must fall. A lower $\bar{\varepsilon}_f$ then implies a higher liquidity premium, thereby making the bubble existence condition (66)

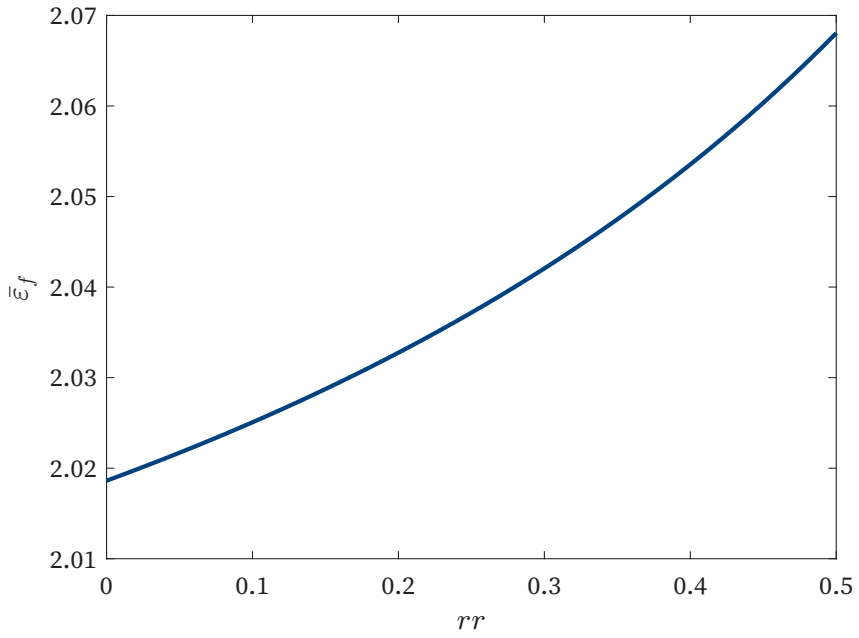


FIGURE 2. Impact of Reserve Requirement on the Bubbleless Investment Threshold

easier to satisfy and the economy more susceptible to the emergence of bubbles. Conversely, forces that boost entrepreneurial net worth or relax credit constraints shift the $\mathcal{H}(\bar{\varepsilon})$ schedule upward, raising $\bar{\varepsilon}_f$ and suppressing the liquidity premium, which ultimately hinders bubble formation or bursts an existing bubble.

As is evident from equation (62), a looser domestic monetary policy stance (higher π) or a looser external financial environment (lower r^*) shifts the $\mathcal{H}(\bar{\varepsilon})$ schedule downward, facilitating the emergence of bubbles. A higher π acts as a heavier inflation tax on banks' reserve holdings, which decreases the net worth of entrepreneurs in real terms and stimulates their demand for bubbles. A lower r^* induces capital inflows, which drives down the domestic real interest rate and fuels a bubble.

The impact of the reserve requirement, rr, however, is analytically ambiguous. On the one hand, the credit rationing channel dictates that a higher rr widens the credit spread, which raises the borrowing threshold $\varepsilon_l(\bar{\varepsilon})$ and decreases the portion of aggregate investment funded by external credit. On the other hand, a higher rr mechanically increases the bank's reserve holdings, which in turn bolsters the entrepreneurial net worth. As these two channels compete, the net displacement of the $\mathcal{H}(\bar{\varepsilon})$ schedule depends on the model parameterization.

To resolve this ambiguity, we resort to a numerical evaluation under our baseline calibration in Section 4.1. Figure 2 plots the resulting equilibrium threshold $\bar{\varepsilon}_f$ as a function of the reserve requirement rr. As the figure illustrates, the net worth channel dominates the adverse effects of credit rationing. A higher rr shifts the $\mathcal{H}(\bar{\varepsilon})$ schedule upward, which increases the fundamental investment threshold $\bar{\varepsilon}_f$, decreasing the liquidity premium of a bubble and hindering its formation.

Long-run Effects of Bubbles. When condition (66) is satisfied, there may exist both a bubbleless and a bubbly equilibrium. As illustrated in equations (45) and (47), the existence of asset bubbles affects aggregate investment along both the intensive and extensive margins. On the one hand, bubbles provide an additional internal funding resource for efficient entrepreneurs to invest along the intensive margin. On the other hand, more inefficient entrepreneurs are encouraged to hold bubbles instead of investing. As a result, we observe $\bar{\varepsilon}_b > \bar{\varepsilon}_f$ as characterized in Proposition 4, which also implies an increased accessible threshold for external funding since $\varepsilon_l(\bar{\varepsilon})$ is strictly increasing in $\bar{\varepsilon}$.

Traditional wisdom [Samuelson (1958), Tirole (1985)] generated from overlapping-generation model predicts that the emergence of bubbles increases the equilibrium risk-free rate. Under such a framework, bubbles act as an alternative vehicle for saving and crowds out investment in physical capital in a production economy so as to increase the marginal product of capital (MPK). Our infinite-horizon model indeed yields the same conclusion regarding the risk-free rate, as the steady-state rate $R(\bar{\varepsilon})$ is strictly increasing in $\bar{\varepsilon}$, but through a fundamentally different mechanism.

In our framework, which features collateral constraints and an endogenous demand for liquidity, a higher investment threshold $\bar{\varepsilon}$ implies a lower liquidity premium for risk-free bonds. This depresses the bond price and drives up its yield. Furthermore, contrary to traditional models, our framework predicts that the MPK in the bubbly equilibrium is actually lower than in the bubbleless equilibrium, as $\frac{r_k}{p_k}(\bar{\varepsilon})$ is strictly decreasing in $\bar{\varepsilon}$ by Lemma 2. This result is intuitive as the aggregate capital stock increases due to the positive intensive margin effect of bubbles on investment, which dominates the negative extensive margin effect illustrated above.

Finally, it is worth noting that the existence of a bubble increases the external debt-to-GDP ratio, as $\frac{b^*}{p_z z}(\bar{\varepsilon})$ is strictly increasing in $\bar{\varepsilon}$. This is also intuitive as a higher domestic risk-free rate naturally induces greater capital inflows.

Bubbly Steady State Comparative Statics. We now conduct a comparative statics analysis to study the long-run impact of domestic policy stances and external financial condition on the bubbly equilibrium.

Figure 3 plots the percentage deviations of several macroeconomic aggregates from their baseline steady state values (y-axis) against an increasing inflation target (x-axis). A higher inflation target exerts a negative long-run effect on the real economy: it raises the size of the bubble but lowers aggregate investment, output, consumption, and labor. As the investment threshold $\bar{\varepsilon}_b$ is anchored by the exogenous sentiment parameter s and discount factor β in equation (67), the real deposit rate remains constant. A higher inflation target then translates proportionally into a higher nominal deposit rate R_b via the Fisher effect.

As discussed above, a heavier inflation tax on bank reserves reduces the net worth of the entrepreneurial sector. To compensate for this loss, credit-constrained entrepreneurs demand more bubble assets to carry wealth forward and fund future investment, thereby growing the size of the bubble. However, due to the increased external finance premium under a higher nominal

rate, fewer entrepreneurs gain access to external credit. This negative extensive margin effect on investment dominates the positive intensive margin effect from the increased bubble size, resulting in a net reduction in aggregate investment, capital accumulation, and output.

Since the real interest rate differential remains invariant, the steady-state external debt-to-output ratio remains constant. Consequently, the absolute level of foreign borrowing falls proportionally with the lower domestic production level. As the domestic economy shrinks, the demand for imported intermediate goods naturally contracts, and exports decline in tandem to preserve the steady state trade balance.

Figure 4 plots the deviations of the variables from their baseline steady state values against an increasing reserve requirement ratio. A higher reserve requirement exerts a contractionary effect on the real economy entirely through the credit rationing channel. As banks must park a larger fraction of their deposit liabilities as reserves while facing an invariant nominal deposit rate, they must pass this increased regulatory cost onto borrowing entrepreneurs to maintain their zero-profit condition. This widens the external finance premium and drives the borrowing cutoff $\bar{\varepsilon}_{lb}$ higher, inducing a negative extensive margin effect that triggers a net reduction in aggregate investment, physical capital, and output.

In addition, a higher reserve requirement alters the composition of the entrepreneurial sector's net worth. The forced increase in reserve holdings absorbs a larger share of the aggregate net worth, effectively crowding out entrepreneurial demand for alternative liquid assets and thereby shrinking the size of the bubble. Combined with the severe drop in physical capital accumulation, this ultimately drives down the aggregate stock price.

Figure 5 plots the comparative statics of the variables against an increasing foreign interest rate. Unlike domestic policy shocks, an increase in the foreign interest rate affects the long-run equilibrium exclusively through the capital account and the resulting trade adjustments. A higher foreign interest rate narrows the real interest rate differential between the domestic economy and the rest of the world. This diminishes the appeal of domestic assets, triggering a decline in capital inflows and driving down the steady-state external debt-to-output ratio. To compensate for this capital outflow, the required improvement in the trade balance forces a relative increase in exports and a contraction in imported intermediate goods. To meet the higher external demand for domestic goods, the production sector expands, which induces higher labor hours and an increase in physical capital investment.

Furthermore, this expansion of physical capital accumulation absorbs a larger share of entrepreneurial net worth. This effectively crowds out entrepreneurial demand for the bubble asset, causing the size of the bubble—and consequently, the aggregate stock price—to shrink.

4. Impulse Responses

This section examines the dynamic responses of the model economy to various shocks. Our analysis focuses on the local dynamics around the bubbly steady state. To conduct this quantitative analysis,

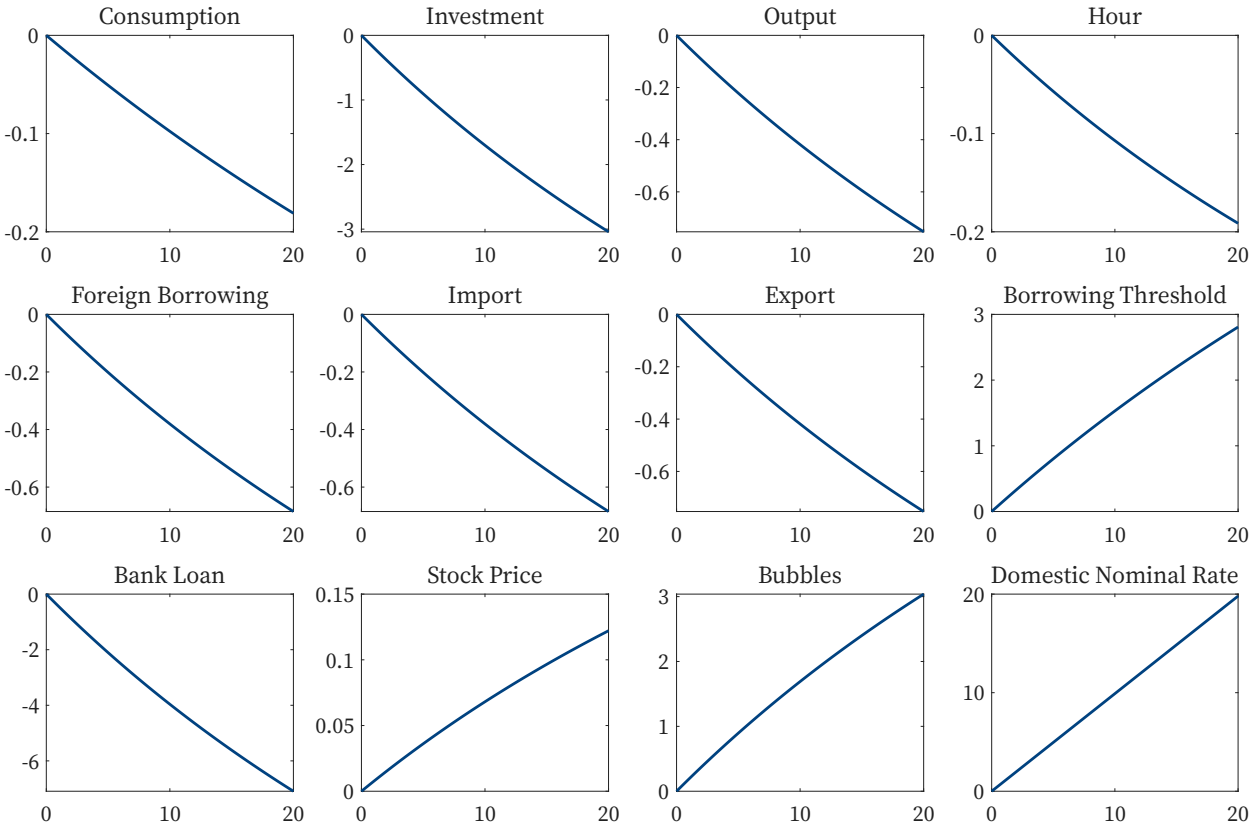


FIGURE 3. Comparative Statics of the Bubbly Steady State: Inflation Target

Note: The vertical axes display the percentage deviation of each variable from its baseline bubbly steady-state value. The horizontal axis represents the percentage deviation of the inflation target π from its baseline calibration given in Section 4.1.

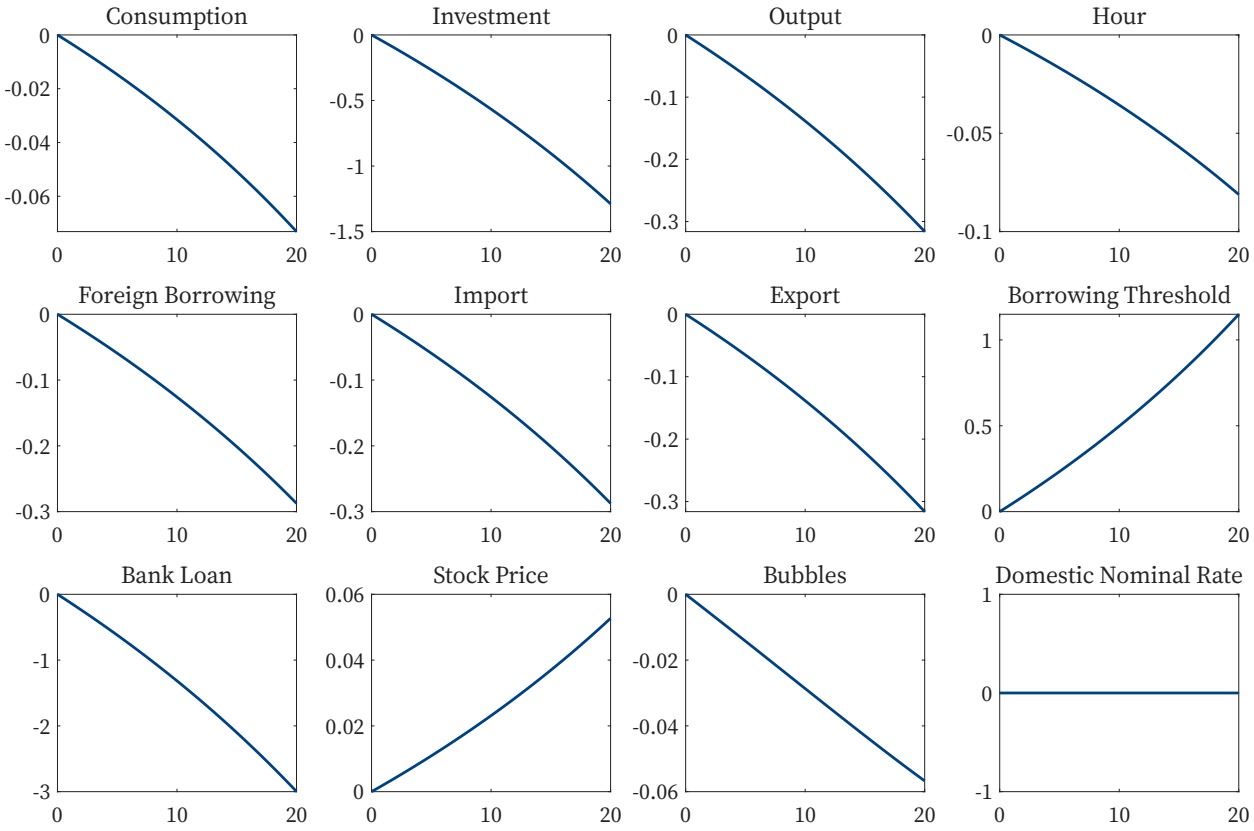


FIGURE 4. Comparative Statics of the Bubbly Steady State: Reserve Requirement

Note: The vertical axes display the percentage deviation of each variable from its baseline bubbly steady-state value. The horizontal axis represents the percentage deviation of the reserve requirement rr from its baseline calibration given in Section 4.1.

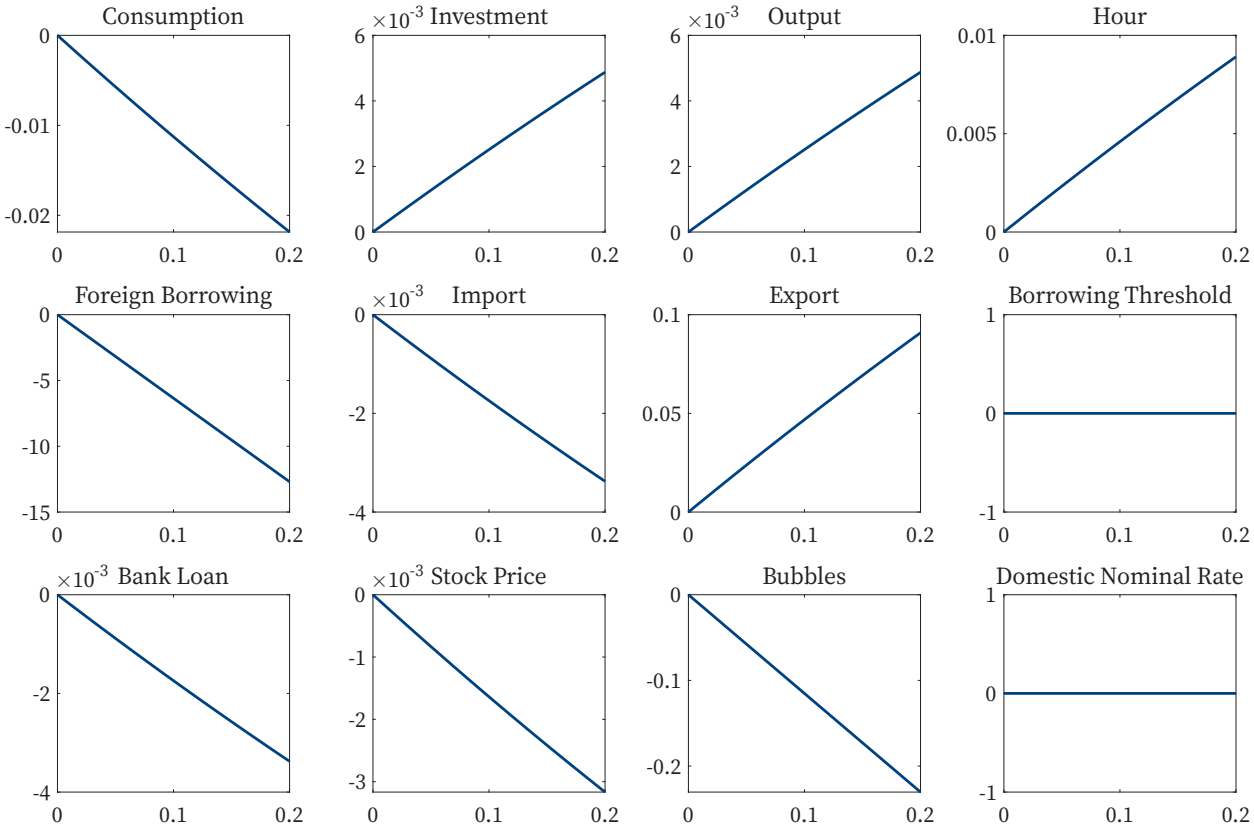


FIGURE 5. Comparative Statics of the Bubbly Steady State: Foreign Interest Rate

Note: The vertical axes display the percentage deviation of each variable from its baseline bubbly steady-state value. The horizontal axis represents the percentage deviation of the exogenous foreign interest rate r^* from its baseline calibration given in Section 4.1.

we first outline the parameter calibration.

4.1. Calibration

The model is calibrated at a quarterly frequency, with parameters chosen to match the long-run features of an abstract small open emerging economy following the literature. The subjective discount factor β is set to 0.98, the physical capital depreciation rate δ is set to 0.05, and the steady-state growth rate g is set to 0.025. For the production technology, the physical capital share α is set to 0.33, and the imported intermediate input share γ is 0.22. The investment adjustment cost parameter Ω_k is set to 2.18. The habit formation parameter h is set to 0.56, and the inverse Frisch elasticity of labor supply φ is set to 2. The weight on labor disutility κ is chosen such that households spend exactly one-third of their time endowment working in the steady state.

Regarding the idiosyncratic investment efficiency shock ε, we assume that it follows a Pareto distribution with the cumulative distribution function $F(\varepsilon) = 1 - \left(\frac{\varepsilon_{\min}}{\varepsilon}\right)^{\eta}$. The shape parameter η governs the dispersion of the distribution and is calibrated to target a steady-state aggregate investment-to-output ratio of 20%. To ensure that the unconditional mean of the investment efficiency shock is unity, the lower bound $\varepsilon_{\min}$ is set to $1 - \eta^{-1}$.

To calibrate the financial friction, the pledgeability parameter μ, is set to 0.30. The steady-state reserve requirement ratio rr is anchored at 0.15. For the external sector, the steady-state foreign interest rate r^* is set to 1.007, and the elasticity of the country-risk premium Ω is calibrated to 0.33 to ensure that external debt accounts for 40% of the banking sector's total debt in the steady state. Determining the appropriate size of asset bubbles in the long-run economy is empirically challenging. In this paper, we set the deterministic steady-state sentiment parameter s to 0.9975, so that newly emerged bubbles account for 0.25% of the aggregate bubble index in each period.

Regarding nominal rigidities, we assume symmetry between the domestic and export retail markets. We set the Calvo price-stickiness parameter to 0.75, ensuring that the average duration of an individual retail price is four quarters. The elasticities of substitution is set to 11, implying a steady-state gross markup of 1.1.

For central bank behavior, we set the long-run inflation target to $\pi = 1.01.$, corresponding to an long-run annual inflation of 4%. To illustrate the short-run model dynamics under different policy regimes, in this section we assume standard Taylor rule responses of $\phi_\pi = 1.5$ and $\phi_z = 0.5$ across all three regimes; whenever the financial stabilization terms are active, we set $\phi_f = 0.5$ and $\phi_{rr} = 5$.

Finally, for the exogenous shock processes, we set the persistence parameters of the shocks to 0.95. The standard deviations of the shocks are calibrated as follows: the short-run productivity shock is set to $\sigma_a = 1.5\%$, the foreign interest rate shock to $\sigma_{r^*} = 0.3\%$, and the sentiment bubble shock to $\sigma_s = 0.15\%$. Table 1 summarizes the calibrated parameter values.

TABLE 1. Parameter Calibration

Parameter	Value	Description
β	0.98	Subjective discount factor
h	0.56	Habit formation in consumption
φ	2.00	Inverse Frisch elasticity of labor supply
κ	17.66	Labor disutility parameter
η	2.11	Pareto distribution shape parameter
$\varepsilon_{\min}$	0.53	Lower bound of the Pareto distribution
α	0.33	Capital share in production
γ	0.22	Imported input share in production
δ	0.05	Capital depreciation rate
Ω_k	2.18	Investment adjustment cost parameter
g	0.025	Long-run TFP growth rate
μ	0.30	Pledgeability parameter (LTV ratio)
rr	0.15	Steady-state reserve requirement ratio
r^*	1.007	Steady-state gross foreign interest rate
Ω	0.33	Debt-elastic country-risk premium parameter
χ, χ^*	0.75	Calvo price rigidity
ψ, ψ^*	11	Elasticity of substitution between differentiated goods
π	1.01	Long-run gross inflation target
ϕ_π	1.50	Taylor rule coefficient on inflation
ϕ_z	0.5	Taylor rule coefficient on output
ϕ_f	0.5	Taylor rule coefficient on asset prices
ϕ_{rr}	5	Reserve requirement reaction to asset prices
s	0.9975	Steady-state fraction of pre-existing bubbles
$\rho_a, \rho_{r^*}, \rho_s$	0.95	Persistence of exogenous shocks
σ_a	1.5	Standard deviation of short-run TFP shock
σ_{r^*}	0.5	Standard deviation of foreign interest rate shock
σ_s	0.15	Standard deviation of sentiment bubble shock

4.2. Short-run Technology shock

Figure 6 shows the impulse responses of several key variables to a one-standard-deviation negative short-run TFP shock across the three policy regimes. We first examine the transmission mechanism under the baseline Taylor Rule (TR) regime. A negative technology shock reduces aggregate output and consumption while driving up real marginal costs for wholesale goods, thereby exerting upward pressure on domestic inflation. Due to price stickiness in both domestic and export markets, aggregate demand adjusts sluggishly, prompting firms to initially increase labor demand to offset the productivity shortfall.

Following the standard Taylor rule, the central bank responds to this inflationary pressure by raising the domestic nominal interest rate. Because bank lending rates are tied to deposit rates via the reserve requirement, the higher policy rate widens the external finance premium, thereby reducing bank lending to the entrepreneurial sector along the extensive margin.

Aggregate investment contracts through both the extensive and intensive margins. On the

extensive margin, the depressed marginal value of capital raises the investment efficiency threshold $\bar{\varepsilon}_t$. On the intensive margin, this higher threshold lowers the liquidity premium, shrinking the bubble's value. Combined with falling aggregate capital rents, this depletes entrepreneurial net worth and reduces the internal funds available to efficient entrepreneurs. As established by Dong, Jiao, and Wang (2025) and Dong, Mei, and Xiao (2025), the presence of an asset bubble introduces a "bubble multiplier" effect: the shrinking bubble suppresses investment and reduces aggregate demand, which further depresses the capital rental rate and Tobin's Q. This pushes $\bar{\varepsilon}_t$ even higher, further compressing the liquidity premium and eroding both the bubble's value and entrepreneurial net worth. Ultimately, concurrent drops in the fundamental value of physical capital and the bubble component drive a severe decline in the aggregate stock price index.

In the external sector, as domestic goods become more expensive due to the decline in productivity, aggregate export demand is depressed and the volume of imported materials required for production increases. This deteriorates the trade balance and expands the stock of foreign borrowing. The dynamics of the nominal exchange rate are dictated by the UIP condition, which acts as an international no-arbitrage condition. As the central bank raises the nominal interest rate to combat domestic inflation, it creates a positive interest rate differential against the rest of the world, so that the domestic currency must be expected to depreciate to balance the debt-elastic foreign interest rate with the expected domestic deposit return in terms of foreign currency.

We now observe the differences in model dynamics when the alternative financial stabilization tools are introduced. When the Augmented Taylor Rule (ATR) regime is active, the central bank faces a policy tradeoff that is exacerbated by the open-economy setting. Because the negative TFP shock collapses asset prices, the ATR mandate instructs the central bank to ease policy compared to the standard Taylor rule, exerting downward pressure on the nominal interest rate. However, as dictated by the UIP condition, this relative monetary easing induces capital outflows and a sharp nominal currency depreciation, which results in more severe imported inflation and ultimately forces the central bank to hike the nominal rate even higher than in the TR regime. This leads to a classic perverse outcome: asset prices suffer a deeper contraction while inflation is more volatile.

In contrast, the MACRO regime circumvents this exchange rate trap by decoupling the policy instruments. The central bank maintains the necessary policy rate hike to anchor inflation expectations and prevent severe currency depreciation, while simultaneously easing the reserve requirement to support bank lending. By reducing the external finance premium, the MACRO regime cushions the investment contraction along the extensive margin. As the investment efficiency threshold $\bar{\varepsilon}_t$ is prevented from rising as sharply, asset prices are stabilized and financial accelerator effects are alleviated without sacrificing nominal stability.

4.3. Foreign Interest Rate Shock

Figure 7 shows the impulse responses of the economy to a one-standard-deviation positive foreign interest rate shock. An unexpected tightening of global financial conditions induces capital outflows and exerts depreciation pressure on the nominal exchange rate. This currency depreciation

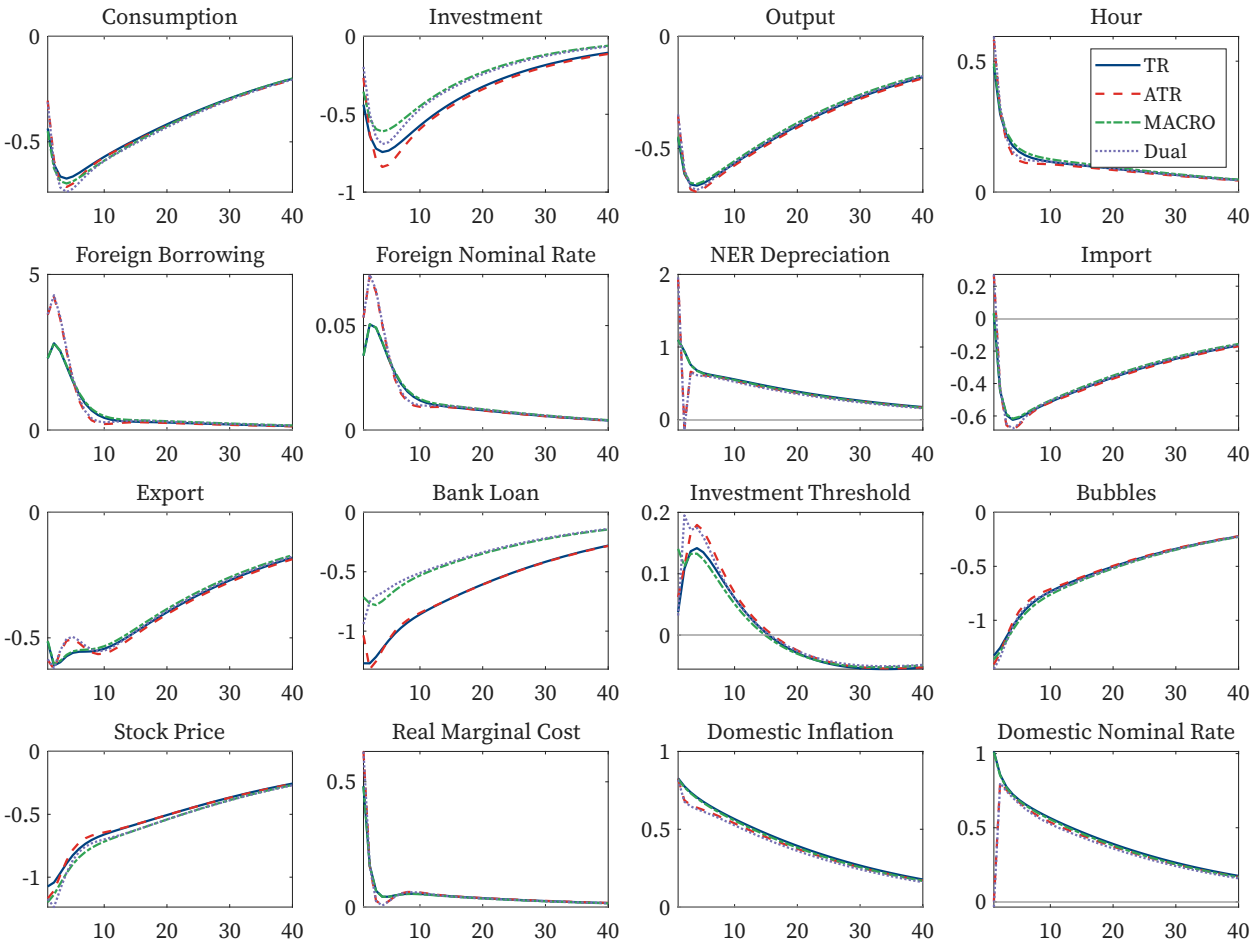


FIGURE 6. Impulse Responses to Short-run TFP Shock

Notes: The figure plots the impulse responses of several variables to a one-standard-deviation adverse short-run productivity shock. All variables are normalized by the nominal exchange rate where applicable, detrended, and expressed as percentage deviations from their respective bubbly steady states. The dynamics are compared across three monetary and macroprudential policy regimes: Taylor Rule regime (TR, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0, ϕ_{rr} = 0), the Augmented Taylor Rule regime (ATR, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0.5, ϕ_{rr} = 0), and the Macroprudential Policy regime (MACRO, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0, ϕ_{rr} = 5).

improves the trade balance by stimulating exports and curbing import demand, but it also raises the domestic currency price of imported intermediate materials. The resulting increase in production costs drives up real marginal costs for domestic firms, generating imported inflationary pressures across the economy.

The central bank counters this imported inflation by raising the domestic policy rate, which reduces domestic aggregate demand. Investment by entrepreneurs contracts along both the extensive and intensive margins. It is worth noting that the decline in the asset bubble and overall stock prices under a positive foreign interest rate shock is significantly larger than that observed in Figure 6 under a negative short-run TFP shock. This outsized response stems from the direct transmission of global financial conditions into domestic asset valuations. As dictated by the UIP condition, an increase in the foreign interest rate drives up the domestic real rate, which directly acts as the discount factor for both the fundamental value of capital (Tobin's Q) and the bubble asset, as shown in Equations (18) and (40). This pronounced decline in asset prices triggers a magnified financial accelerator mechanism: the drop in Tobin's Q raises the investment efficiency threshold $\bar{\varepsilon}_t$, suppressing investment along the extensive margin; the contraction of the bubble asset depletes entrepreneurial net worth, activating the "bubble multiplier" effect discussed previously.

Under the ATR regime, the central bank's attempt to stabilize the stock price slump via the policy rate once again backfires. By exerting downward pressure on the domestic interest rate, the ATR rule exacerbates the initial currency depreciation and amplifies imported inflationary pressures. The MACRO regime instead maintains the necessary nominal policy rate hike to defend the currency, while simultaneously injecting liquidity into the banking sector to dampen the fall in investment and domestic aggregate demand, effectively isolating the domestic credit cycle from global financial tightening.

4.4. Sentiment Bubble Shock

Figure 8 illustrates the impulse responses of the economy to a one-standard-deviation adverse sentiment bubble shock. A negative sentiment shock immediately destroys a fraction of pre-existing bubbles, triggering a collapse in the aggregate bubble index and total stock market capitalization. Because the bursting of the bubble reduces entrepreneurial net worth, entrepreneurs are forced to rely more heavily on external credit to finance investment, causing a spike in the demand for domestic bank loans. However, as firms remain subject to strict collateral constraints, aggregate investment contracts along the intensive margin. Interestingly, as illustrated in Dong, Miao, and Wang (2020), this investment contraction is partially mitigated by a general equilibrium feedback loop on the extensive margin: as physical capital becomes scarcer, its fundamental valuation rises, which then lowers the idiosyncratic investment efficiency threshold $\bar{\varepsilon}_t$ and allows a larger mass of entrepreneurs to participate in investment.

Overall, the loss of entrepreneurial financial wealth drives a contraction in domestic aggregate demand, depressing both aggregate output and consumption, while imposing downward pressure on domestic inflation and prompting the central bank to cut the domestic nominal policy rate

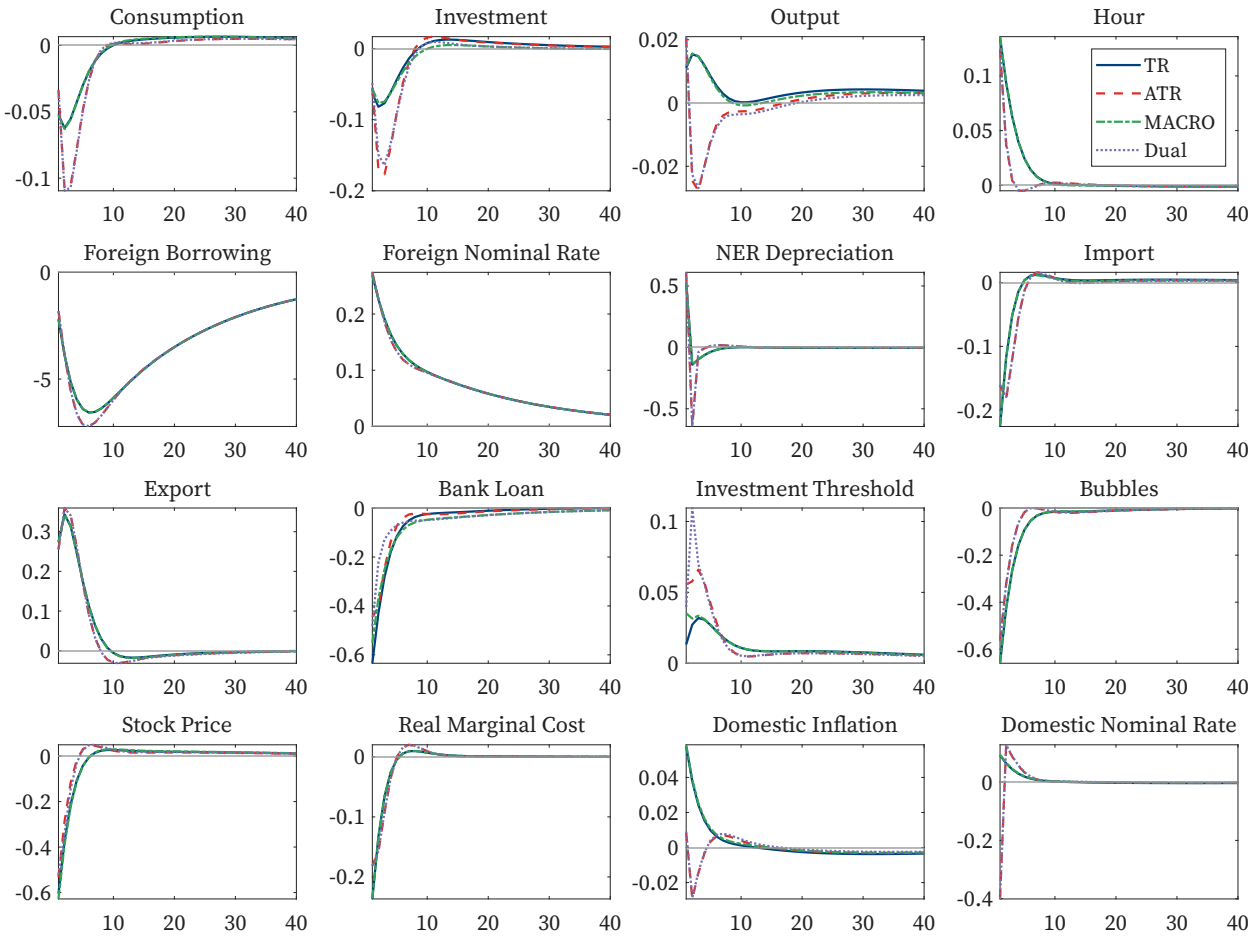


FIGURE 7. Impulse Responses to Foreign Interest Rate Shock

Note: The figure plots the impulse responses of several variables to a one standard deviation positive foreign interest rate shock. All variables are normalized by the nominal exchange rate where applicable, detrended, and expressed as percentage deviations from their respective bubbly steady states. The dynamics are compared across three monetary and macroprudential policy regimes: Taylor Rule regime (TR, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0, ϕ_{rr} = 0), the Augmented Taylor Rule regime (ATR, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0.5, ϕ_{rr} = 0), and the Macroprudential Policy regime (MACRO, ϕ_π = 1.5, ϕ_z = 0.5, ϕ_f = 0, ϕ_{rr} = 5).

accordingly.

In the external sector, the drop in domestic wholesale prices makes domestic goods relatively cheaper on the global market, stimulating export demand. Simultaneously, the contraction in total wholesale production reduces the required volume of imported intermediate materials. Consequently, the small open economy runs a temporary trade surplus, which decreases the stock of foreign borrowing and triggers nominal exchange rate adjustments governed by the UIP condition to maintain international arbitrage.

The difference in the nominal exchange rate dynamics across the policy regimes is more pronounced in this case. Under the ATR regime, the mandate to stabilize the stock price dictates an aggressive policy rate cut, which triggers an immediate nominal currency depreciation. This stokes imported inflation and forces the central bank to hike the nominal rate on impact. Consequently, the ATR rule fails to deliver the intended monetary easing, leaving the stock market vulnerable. The MACRO regime, by contrast, achieves effective stabilization through the banking channel. A sharp reduction in the reserve requirement compresses the external finance premium, ensuring that entrepreneurs can access the credit necessary to offset their lost financial wealth. Because this intervention operates directly on the financial friction rather than the nominal interest rate, it entirely avoids the disruptive exchange rate depreciation.

5. Welfare Evaluation of Alternative Policy Regimes

Having investigated the model dynamics facing different shocks under alternative policy regimes, we now adopt a normative perspective. In this section, we search for the optimal parameter values in the monetary policy rule (37) and the macroprudential policy rule (39) across our three policy regimes by evaluating the resulting household welfare. Following Schmitt-Grohé and Uribe (2007) and Faia and Monacelli (2007), our criterion is the conditional expectation of the representative household's lifetime utility, computed using a second-order approximation. Specifically, we assume the economy resides at its deterministic bubbly steady state in period −1. From period 0 onward, the economy is subjected to exogenous shocks, and the central bank implements policy rules to stabilize the economy.

To examine whether—and how—the central bank should respond to asset prices, we evaluate the optimal Augmented Taylor Rule (ATR) and Macroprudential Policy (MACRO) regimes by comparing their relative welfare gains against the optimized Taylor Rule (TR) baseline. Let Wel^*_{TR} denote the welfare under this optimized TR baseline, and let Wel^* denote the welfare under the alternative optimized policy regime. We represent the welfare gain Δ by the Consumption Equivalent Variation (CEV), defined as the constant fraction of additional consumption required under the baseline regime to render the household indifferent between the baseline and the alternative policy regime. Given the household's utility function, Δ satisfies the following condition:

$$E_{-1}\sum_{t=0}^{\infty}\beta^t\left[\ln\left((1+\Delta)(C_t-hC_{t-1})\right)-\kappa\frac{N_t^{1+\varphi}}{1+\varphi}\right]=Wel^*,$$

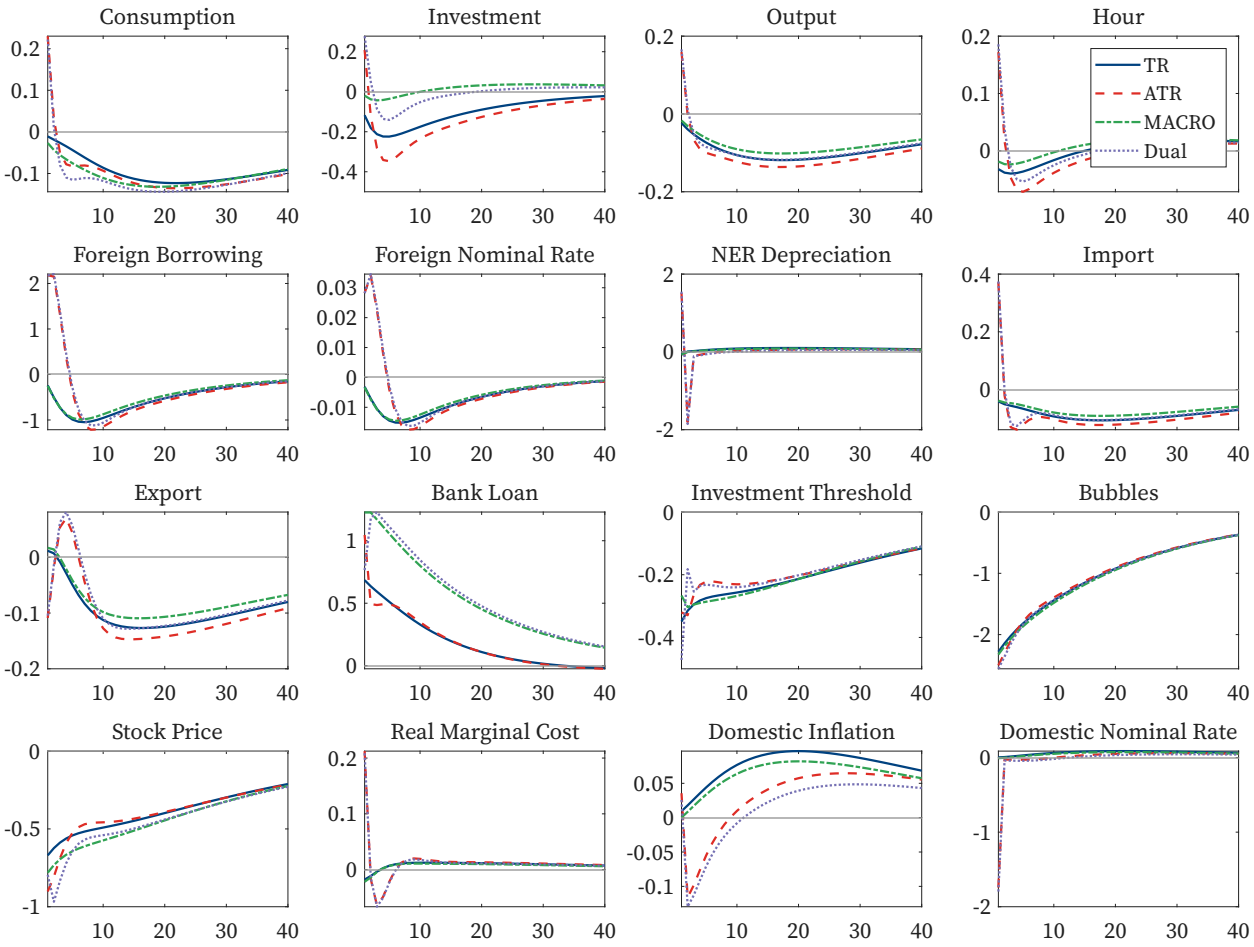


FIGURE 8. Impulse Responses to Sentiment Bubble Shock

Note: The figure plots the impulse responses of several variables to a one standard deviation negative sentiment bubble shock. All variables are normalized by the nominal exchange rate where applicable, detrended, and expressed as percentage deviations from their respective bubbly steady states. The dynamics are compared across four monetary and macroprudential policy regimes: The dynamics are compared across three monetary and macroprudential policy regimes: Taylor Rule regime (TR, $\phi_\pi = 1.5$, $\phi_z = 0.5$, $\phi_f = 0$, $\phi_{rr} = 0$), the Augmented Taylor Rule regime (ATR, $\phi_\pi = 1.5$, $\phi_z = 0.5$, $\phi_f = 0.5$, $\phi_{rr} = 0$), and the Macroprudential Policy regime (MACRO, $\phi_\pi = 1.5$, $\phi_z = 0.5$, $\phi_f = 0$, $\phi_{rr} = 5$).

TABLE 2. Optimal Simple Rules and Welfare Gains

Policy Regime	ϕ_π	ϕ_z	ϕ_f	ϕ_{rr}	Δ %
All Shocks					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.0	–	0.0000
MACRO	3.0	0.0	–	10.0	0.0918
Technology Shock Only					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.0	–	0.0000
MACRO	3.0	0.0	–	10.0	0.0285
Foreign Interest Rate Shock Only					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.1	–	0.0000
MACRO	3.0	0.0	–	0.0	0.0000
Sentiment Bubble Shock Only					
TR	1.4	0.0	–	–	0.0000
ATR	1.4	0.0	0.1	–	0.0001
MACRO	1.1	0.0	–	10.0	0.0798

Notes: TR denotes the standard Taylor Rule, ATR denotes the Augmented Taylor Rule targeting the financial spread (ϕ_f), and MACRO denotes the coordinated regime utilizing the reserve requirement (ϕ_{rr}). The CEV Gain (Δ) represents the permanent percentage increase in baseline consumption required to match the welfare of the respective optimized regime.

where $\{C_t\}$ and $\{N_t\}$ denote the state-contingent equilibrium paths of consumption and labor hours, respectively, under the baseline optimized TR regime. By definition, these baseline allocations satisfy:

$$E_{-1}\sum_{t=0}^{\infty}\beta^t\left[\ln(C_t - hC_{t-1}) - \kappa\frac{N_t^{1+\varphi}}{1+\varphi}\right] = Wel^*_{TR}.$$

Table 2 reports the optimal parameter values for the policy rules and the associated welfare gains. To provide a comprehensive and operationally realistic recipe for the central bank, we first evaluate the optimal policy parameters when the economy is subjected to all shocks simultaneously. Subsequently, we decompose the results by evaluating each shock individually to understand the specific economic mechanisms driving the central bank's optimal response.

As Table 2 illustrates, under the composite scenario encompassing all shocks, the MACRO regime strictly dominates. The optimal response to asset price fluctuations via the policy rate (ϕ_f) in the ATR regime falls to zero, yielding no welfare improvement over the baseline TR. In stark contrast, the MACRO regime optimally deploys an aggressive countercyclical reserve requirement ($\phi_{rr} = 10.0$), generating a substantial welfare gain equivalent to a **0.0918

Decomposing the welfare gains by individual shocks reveals the mechanisms driving this divergence. When the economy faces internal, domestic disturbances—such as technology or sentiment bubble shocks—the MACRO regime is highly active and highly effective. Consider a negative sentiment shock, which directly destroys entrepreneurial net worth. Adjusting the interest rate to stabilize the financial sector (the ATR regime) creates a severe conflict with the central bank's exchange rate and inflation mandates, yielding a negligible **0.0001

Conversely, the optimal policy landscape shifts dramatically when the economy is hit by a foreign interest rate shock. Under this specific scenario, the optimal macroprudential coefficient (ϕ_{rr}) drops to zero, and the standard Taylor Rule proves to be the robust baseline. An unexpected tightening of global financial conditions induces capital outflows and currency depreciation, making it fundamentally an external pricing shock rather than an internal credit friction distortion. Attempting to inject domestic liquidity via reserve requirements fails to address the uncovered interest parity gap and capital flight. Instead, the central bank must rely purely on the conventional policy rate to defend the currency, stem outflows, and anchor imported inflation.

6. Leaning Against Bubbles?

A persistent debate in the optimal monetary policy literature concerns whether central banks should "lean against the wind" of asset price booms. A standard critique of such policies is an informational friction: in practice, central banks cannot easily disentangle efficient, fundamental-driven asset price appreciation from inefficient, speculative bubble expansions. Consequently, responding to aggregate asset prices risks inadvertently stifling fundamental economic growth.

To address this critique, we conduct a theoretical robustness check. We assume the central bank possesses perfect information and is entirely capable of isolating the bubble from fundamentals. We modify the Augmented Taylor Rule (ATR) so that the monetary authority leans directly against the pure rational bubble component rather than the aggregate stock price index. If the failure of the ATR in our baseline results was purely due to the inability to distinguish fundamentals from bubbles, targeting the bubble directly should yield positive welfare gains and a highly active policy coefficient ($\phi_f > 0$).

Table 3 reports the optimized parameters and welfare outcomes under this alternative specification. Strikingly, the results demonstrate that even with perfect information, the optimal monetary response to the bubble (ϕ_f) falls entirely to zero for the composite shock, the technology shock, and the sentiment shock. The ATR fails to generate any welfare improvement over the standard Taylor Rule (TR) baseline. Meanwhile, the MACRO regime remains perfectly robust, retaining its aggressive reserve requirement stance ($\phi_{rr} = 10.0$) and preserving its substantial welfare gains across the board.

This finding yields a profound theoretical implication. It proves that the suboptimality of using the interest rate to lean against the wind is not merely a consequence of informational constraints or the inability to parse fundamentals from speculation. Instead, the short-term interest rate is

TABLE 3. Robustness Check: Optimal Simple Rules Targeting the Bubble

Policy Regime	ϕ_π	ϕ_z	ϕ_f	ϕ_{rr}	Δ %
All Shocks (Composite Baseline)					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.0	–	0.0000
MACRO	3.0	0.0	–	10.0	0.0918
Technology Shock Only					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.0	–	0.0000
MACRO	3.0	0.0	–	10.0	0.0285
Foreign Interest Rate Shock Only					
TR	3.0	0.0	–	–	0.0000
ATR	3.0	0.0	0.1	–	0.0000
MACRO	3.0	0.0	–	0.0	0.0000
Sentiment Bubble Shock Only					
TR	1.4	0.0	–	–	0.0000
ATR	1.4	0.0	0.0	–	0.0000
MACRO	1.1	0.0	–	10.0	0.0798

Notes: This table presents the optimal policy parameters when the Augmented Taylor Rule (ATR) targets the rational bubble directly rather than the aggregate asset price. TR denotes the standard Taylor Rule, and MACRO denotes the coordinated regime utilizing the reserve requirement (ϕ_{rr}). The CEV Gain (Δ) represents the permanent percentage increase in baseline consumption required to match the welfare of the respective optimized regime.

fundamentally the wrong tool for the job. Because the interest rate is a blunt macroeconomic instrument, attempting to pop or restrain a bubble by tightening monetary policy inflicts excessive collateral damage on the productive sectors of the real economy and disrupts the exchange rate. By contrast, macroprudential policy acts as a surgical instrument. Deploying the reserve requirement strictly targets the over-leveraged financial sector, successfully insulating the economy from bubble-driven volatility without compromising the central bank's primary macroeconomic objectives.

7. Concluding Remarks

We have developed a Dynamic New Keynesian (DNK) model for a small open economy that incorporates rational asset bubbles and banking frictions. By integrating heterogeneous, credit-constrained entrepreneurs with financial intermediaries subject to reserve requirements, our framework provides a comprehensive laboratory to evaluate the optimal design and coordination of monetary and macroprudential policies. This approach bridges the established financial accelerator mechanism with contemporary theories of asset bubbles, allowing us to examine how external and internal

shocks propagate through the domestic financial system.

Our steady-state analysis demonstrates that the emergence and size of asset bubbles are intricately tied to both global financial conditions and domestic policy stances. Specifically, a looser external financial environment, driven by lower foreign interest rates, induces capital inflows and depresses domestic real interest rates, thereby fueling domestic asset bubbles. Conversely, tightening macroprudential policy via higher reserve requirements absorbs a larger share of aggregate net worth, which crowds out entrepreneurial demand for bubble assets and restricts their formation.

Dynamically, we find that asset bubbles act as a powerful multiplier for exogenous macroeconomic disturbances. When global financial conditions tighten, the collapse of domestic bubbles destroys corporate collateral and exacerbates the economic downturn. Through a welfare evaluation of alternative policy regimes, we demonstrate the superiority of deploying active macroprudential policy—specifically, countercyclical reserve requirements—to stabilize the domestic credit cycle. Crucially, we show that utilizing the standard policy interest rate to "lean against the wind" of asset price fluctuations is strictly suboptimal. Because the short-term interest rate is a blunt instrument, targeting bubbles directly inflicts excessive collateral damage on the productive real economy and exacerbates exchange rate volatility. Instead, a coordinated regime that allows monetary policy to focus purely on nominal stability, while delegating financial stabilization to targeted macroprudential tools, yields substantial welfare improvements.

In an era characterized by elevated geoeconomic fragmentation and heightened capital flow volatility, emerging and developing economies face increasingly complex policy trade-offs. Our analysis underscores that conventional monetary policy frameworks alone may be insufficient to maintain macroeconomic stability when facing the dual threats of asset bubbles and external financial shocks. By illustrating the efficacy of macroprudential tools in insulating the domestic economy from global financial cycles, our findings provide a theoretical foundation for more resilient policy coordination. Future research could fruitfully extend this framework to explore the implications of sovereign debt dynamics, foreign exchange interventions, or targeted capital controls in further mitigating the spillover effects of global fragmentation.

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Appendix A. Proof of Propositions in Section 2

A.1. Proof of Proposition 1

PROOF. We conjecture that the entrepreneur's value function takes the linear form:

$$V_{jt}(\varepsilon_{jt}, K_{j,t-1}, B_{j,t-1}, L_{j,t-1}, \{H_{j,t-1|t-k}\}_{k=0}^{\infty}) = \phi_t^k(\varepsilon_{jt})K_{j,t-1} + \phi_t^b(\varepsilon_{jt})B_{j,t-1} - \phi_t^l(\varepsilon_{jt})L_{j,t-1} + \sum_{k=0}^{\infty} \phi_{t|t-k}^h(\varepsilon_{jt})H_{j,t-1|t-k},$$

where the coefficients $\phi_t^i(\varepsilon_{jt})$ are to be determined. We define the expected marginal valuations as:

$$Q_t = \mathbb{E}_t\beta\frac{\Lambda_{t+1}}{\Lambda_t}\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \phi_{t+1}^k(\varepsilon)dF(\varepsilon), \quad \nu_t^b = \mathbb{E}_t\beta\frac{\Lambda_{t+1}}{\Lambda_t}\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \phi_{t+1}^b(\varepsilon)dF(\varepsilon),$$

$$\nu_t^l = \mathbb{E}_t\beta\frac{\Lambda_{t+1}}{\Lambda_t}\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \phi_{t+1}^l(\varepsilon)dF(\varepsilon), \quad \nu_{t|t-k}^h = \mathbb{E}_t\beta\frac{\Lambda_{t+1}}{\Lambda_t}\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \phi_{t+1|t-k}^h(\varepsilon)dF(\varepsilon).$$

Substituting the flow-of-funds constraint (14) and the capital accumulation equation (12) into the Bellman equation (15), the objective function to maximize becomes:

$$Obj = [R_{kt} + Q_t(1-\delta)]K_{j,t-1} + R_{t-1}B_{j,t-1} - R_{l,t-1}L_{j,t-1} + \sum_{k=0}^{\infty} P_{t|t-k}^h(1-\delta_h)H_{j,t-1|t-k}$$

$$+(Q_t\varepsilon_{jt} - P_{kt})I_{jt} + (\nu_t^b - 1)B_{jt} + (1-\nu_t^l)L_{jt} + \sum_{k=0}^{\infty}(\nu_{t|t-k}^h - P_{t|t-k}^h)H_{j,t|t-k}.$$

To ensure finite optimal choices for unconstrained firms, we must have $\nu_t^b \le 1$ and $\nu_{t|t-k}^h \le P_{t|t-k}^h$. In an equilibrium with active trading, these hold with strict equality:

$$\nu_t^b = 1, \quad \nu_{t|t-k}^h = P_{t|t-k}^h.$$

Because external loans and internal deposits are symmetrically structured but subject to different interest rates, it follows that $\nu_t^l = \frac{R_{lt}}{R_t}\nu_t^b = \frac{R_{lt}}{R_t}$. Since the reserve requirement imposes $R_{lt} > R_t$, we have $\nu_t^l > 1$, which makes the coefficient on loans strictly negative ($1 - \frac{R_{lt}}{R_t} < 0$).

Let NW_{jt} denote the firm's available internal net worth (the first line of the objective function). We analyze the firm's optimal choices across three distinct regions of investment efficiency ε_{jt}:

Case 1: $\varepsilon_{jt} \le \bar{\varepsilon}_t \equiv \frac{P_{kt}}{Q_t}$**.** The marginal value of investment is non-positive, thus the firm does not invest ($I_{jt} = 0$). Since borrowing is costly, it does not borrow ($L_{jt} = 0$), and it is indifferent regarding B_{jt} and $H_{j,t|t-k}$. The value function simplifies to $V_{jt} = NW_{jt}$. Matching coefficients yields:

$$\phi_t^k(\varepsilon_{jt}) = R_{kt} + Q_t(1-\delta), \quad \phi_t^b(\varepsilon_{jt}) = R_{t-1}, \quad \phi_t^l(\varepsilon_{jt}) = R_{l,t-1}, \quad \phi_{t|t-k}^h(\varepsilon_{jt}) = P_{t|t-k}^h(1-\delta_h).$$

Case 2: $\bar{\varepsilon}_t < \varepsilon_{jt} \le \bar{\varepsilon}_{lt} \equiv \frac{R_{lt}}{R_t}\bar{\varepsilon}_t$**.** The firm finds it profitable to invest ($I_{jt} > 0$). To maximize investment, it liquidates all internal funds, setting $D_{jt} = 0$, $B_{jt} = 0$, and $H_{j,t|t-k} = 0$. However,

because the marginal return on investment $\frac{Q_t \varepsilon_{jt}}{P_{kt}}$ does not exceed the marginal cost of borrowing $\frac{R_{lt}}{R_t}$, it chooses not to borrow ($L_{jt} = 0$). Substituting $P_{kt} I_{jt} = NW_{jt}$ into the objective function yields $V_{jt} = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} NW_{jt} + Q_t(1-\delta)K_{j,t-1}$. Matching coefficients yields:

$$\phi_t^k(\varepsilon_{jt}) = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} R_{kt} + Q_t(1-\delta), \quad \phi_t^b(\varepsilon_{jt}) = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} R_{t-1}, \quad \phi_t^l(\varepsilon_{jt}) = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} R_{l,t-1}, \quad \phi_{t|t-k}^h(\varepsilon_{jt}) = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} P_{t|t-k}^h (1-\delta_h).$$

Case 3: $\varepsilon_{jt} > \bar{\varepsilon}_{lt}$**.** The firm's efficiency is high enough that the return on investment strictly dominates the cost of external credit. It exhausts internal funds and borrows up to the collateral constraint $L_{jt} = \mu P_{kt} K_{j,t-1}$. Substituting $P_{kt} I_{jt} = NW_{jt} + \mu P_{kt} K_{j,t-1}$ into the objective function yields:

$$V_{jt} = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} NW_{jt} + Q_t(1-\delta)K_{j,t-1} + \left(\frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} - \frac{R_{lt}}{R_t}\right) \mu P_{kt} K_{j,t-1}.$$

Matching coefficients for this region yields:

$$\phi_t^k(\varepsilon_{jt}) = \frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} R_{kt} + Q_t(1-\delta) + \left(\frac{\varepsilon_{jt}}{\bar{\varepsilon}_t} - \frac{R_{lt}}{R_t}\right) \mu P_{kt},$$

while the other coefficients remain identical to Case 2.

Finally, we aggregate these piece-wise coefficients by integrating over the distribution $F(\varepsilon)$. Taking the expected values of $\phi_t^k(\varepsilon_{jt})$, $\phi_t^b(\varepsilon_{jt})$, and $\phi_{t|t-k}^h(\varepsilon_{jt})$, and substituting them back into the definitions for Q_{t-1}, v_{t-1}^b, and $v_{t-1|t-k}^h$ directly recovers the asset pricing equations (18), (19), and (20). □

A.2. Proof of Proposition 2

PROOF. We derive the balance of payments (BOP) condition by consolidating the budget constraints, zero-profit conditions, and market-clearing conditions of all agents in the small open economy.

First, we aggregate the flow-of-funds constraints across all entrepreneurs. By convention, the choice variable for the newly endowed bubble vintage is set to $H_{j,t-1|t} = \frac{\delta_h}{1-\delta_h}$. Thus, the term for the new bubble vintage ($k = 0$) in the summation simplifies to $P_{t|t}^h[\delta_h - H_{j,t|t}]$, correctly capturing the free endowment of δ_h units of new bubbles. Integrating over all entrepreneurs $j \in [0,1]$ and applying the bubble market-clearing condition $\int H_{j,t|t-k}\, dj = \delta_h(1-\delta_h)^k$, the aggregate net cash flow from trading all vintages of bubbles cancels out exactly to zero. The aggregate entrepreneurial dividend simplifies to:

$$D_t = R_{kt} K_{t-1} - P_{kt} I_t + R_{t-1} B_{t-1} - B_t + L_t - R_{l,t-1} L_{t-1}.$$

Next, we substitute the aggregate profits from domestic banks (D_t^b), capital producers (D_t^k), and retailers (D_t^r) into the household's budget constraint (1). Using the bank's flow of funds (6) and the central bank transfer $T_t = RE_t - RE_{t-1}$, the domestic financial positions ($B_{t-1}, L_{t-1}, RE_{t-1}$) cancel

out. The household budget becomes:

$$P_t C_t = W_t N_t + R_{kt} K_{t-1} - P_{kt} I_t - B_t + L_t + RE_t - e_t R^*_{t-1} B^*_{t-1} + D^k_t + D^r_t.$$

From the bank's balance sheet (5), we substitute $L_t + RE_t - B_t = e_t B^*_t$. We also substitute the definitions of capital producer and retailer profits: $D^k_t = P_{kt} I_t - P_t I^{tot}_t$ (where I^{tot}_t includes adjustment costs) and $D^r_t = P_t Y_t - P_{zt} Z_t + e_t \Delta^*_{pt} Y^*_t$. This yields:

$$P_t C_t = W_t N_t + R_{kt} K_{t-1} + e_t B^*_t - e_t R^*_{t-1} B^*_{t-1} - P_t I^{tot}_t + P_t Y_t - P_{zt} Z_t + e_t \Delta^*_{pt} Y^*_t.$$

From the constant returns to scale technology and competitive input pricing of the wholesale producers, total revenues equal total factor payments:

$$P_{zt} Z_t = W_t N_t + R_{kt} K_{t-1} + e_t P^*_t M_t.$$

Using this identity to eliminate the domestic factor payments from the aggregated household budget, we obtain:

$$P_t C_t = -e_t P^*_t M_t + e_t B^*_t - e_t R^*_{t-1} B^*_{t-1} - P_t I^{tot}_t + P_t Y_t + e_t \Delta^*_{pt} Y^*_t.$$

Finally, applying the domestic final goods market-clearing condition (54), $P_t Y_t = P_t C_t + P_t I^{tot}_t$, cancels out the domestic consumption and total investment terms entirely. Rearranging the remaining components gives the required balance of payments condition:

$$e_t \Delta^*_{pt} Y^*_t - e_t P^*_t M_t = e_t R^*_{t-1} B^*_{t-1} - e_t B^*_t.$$

□

Appendix B. Equilibrium System

B.1. Normalized System

- **Households:**

$$\lambda_t = \frac{1}{p_t}\left(\frac{1}{C_t - hC_{t-1}} - \mathbb{E}_t\beta\frac{h}{C_{t+1} - hC_t}\right),$$

$$\lambda_t w_t = \kappa N_t^{\varphi}.$$

- **Banks:**

$$l_t + re_t = b_t + B_t^*,$$

$$re_t = \frac{rr_t}{1 - rr_t} l_t,$$

$$R_{lt} = \frac{R_t - rr_t}{1 - rr_t},$$

$$\mathbb{E}_t\beta\frac{\lambda_{t+1}}{\lambda_t}\left[\frac{R_t}{\pi_{e,t+1}} - R_t^*\right] = \mathbb{E}_t\beta\frac{\lambda_{t+1}}{\lambda_t}\left[\Omega\exp\left(\frac{B_t^*}{p_{zt}Z_t} - \underline{B}^*\right)\frac{B_t^*}{p_{zt}Z_t}\right],$$

$$R_t^* = r_t^* + \Omega\left[\exp\left(\frac{B_t^*}{p_{zt}Z_t} - \underline{B}^*\right) - 1\right].$$

- **Entrepreneurs:**

$$\bar{\varepsilon}_t = \frac{p_{kt}}{q_t},$$

$$\bar{\varepsilon}_{lt} = \frac{R_{lt}}{R_t}\bar{\varepsilon}_t,$$

$$q_t = \mathbb{E}_t\beta\frac{\lambda_{t+1}}{\lambda_t}\Bigg[r_{k,t+1} + (1-\delta)q_{t+1} + r_{k,t+1}\int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1\right)dF(\varepsilon) + \mu p_{k,t+1}\int_{\bar{\varepsilon}_{l,t+1}}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - \frac{R_{l,t+1}}{R_{t+1}}\right)dF(\varepsilon)\Bigg],$$

$$\frac{1}{R_t} = \mathbb{E}_t\beta\frac{\lambda_{t+1}}{\lambda_t\pi_{e,t+1}}\left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1\right)dF(\varepsilon)\right],$$

$$p_t^h = \mathbb{E}_t\beta\frac{\lambda_{t+1}}{\lambda_t}p_{e,t+1}^h\left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}}\left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1\right)dF(\varepsilon)\right],$$

$$p_{et}^{h} = s_t p_t^{h},$$

$$l_t = \mu p_{kt} K_{t-1}\big(1 - F(\bar{\varepsilon}_{lt})\big),$$

$$nw_t = r_{kt} K_{t-1} + \frac{re_{t-1}}{\pi_{et}} - \frac{R_{t-1}}{\pi_{et}} B_{t-1}^{*} + p_t^{h},$$

$$I_t = \frac{1}{p_{kt}}\left[nw_t\big(1 - F(\bar{\varepsilon}_t)\big) + l_t\right],$$

$$K_t = (1-\delta)K_{t-1} + \left(I_t - \frac{l_t}{p_{kt}}\right)\frac{\int_{\bar{\varepsilon}_t}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_t)} + \frac{l_t}{p_{kt}}\frac{\int_{\bar{\varepsilon}_{lt}}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_{lt})},$$

$$p_{st} = q_t K_t + \frac{re_t}{R_t} - B_t^{*} + p_t^{h}.$$

- **Wholesale Goods Producers:**

$$A_t = A_t^{g} \exp(a_t),$$

$$Z_t = K_{t-1}^{\alpha}(A_t N_t)^{1-\alpha-\gamma} M_t^{\gamma},$$

$$r_{kt} = \alpha \frac{p_{zt} Z_t}{K_{t-1}},$$

$$w_t = (1-\alpha-\gamma)\frac{p_{zt} Z_t}{N_t},$$

$$M_t = \gamma p_{zt} Z_t.$$

- **Capital Producers:**

$$p_{kt} = \left[1 + \frac{\Omega_k}{2}\left(\frac{I_t}{I_{t-1}} - \exp(g)\right)^2\right] p_t + \Omega_k p_t \left(\frac{I_t}{I_{t-1}} - \exp(g)\right)\frac{I_t}{I_{t-1}}$$
$$- \Omega_k \mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} p_{t+1}\left(\frac{I_{t+1}}{I_t} - \exp(g)\right)\left(\frac{I_{t+1}}{I_t}\right)^2.$$

- **Retailers:**

$$\pi_t = \pi_{et}\frac{p_t}{p_{t-1}},$$

$$p_t = \left[(1-\chi)\tilde{p}_{rt}^{1-\psi} + \chi \pi_{t-1}^{1-\psi}\left(\frac{p_{t-1}}{\pi_{et}}\right)^{1-\psi}\right]^{\frac{1}{1-\psi}},$$

$$\tilde{p}_{rt} = \frac{\psi}{\psi - 1}\frac{\gamma_{nt}}{\gamma_{dt}},$$

$$\gamma_{nt} = \lambda_t p_{zt} p_t^{\psi} Y_t + \chi\beta\pi_t^{-\psi}\mathbb{E}_t\pi_{e,t+1}^{\psi}\gamma_{n,t+1},$$

$$\gamma_{dt} = \lambda_t p_t^{\psi} Y_t + \chi\beta\pi_t^{1-\psi}\mathbb{E}_t\pi_{e,t+1}^{\psi-1}\gamma_{d,t+1},$$

$$\tilde{P}_{rt}^{*} = \frac{\psi^{*}}{\psi^{*} - 1}\frac{\Gamma_{nt}^{*}}{\Gamma_{dt}^{*}},$$

$$\Gamma_{nt}^{*} = \lambda_t p_{zt} Y_t^{*} + \chi^{*}\beta\mathbb{E}_t\Gamma_{n,t+1}^{*},$$

$$\Gamma_{dt}^{*} = \lambda_t Y_t^{*} + \chi^{*}\beta\mathbb{E}_t\Gamma_{d,t+1}^{*}.$$

- **Central Bank:**

$$\ln\left(\frac{R_t}{R}\right) = \phi_\pi \ln\left(\frac{\pi_t}{\pi}\right) + \phi_z \ln\left(\frac{Y_t}{Y}\right) + \phi_f \ln\left(\frac{p_{st}/p_t}{p_s/p}\right),$$

$$rr_t - rr = \phi_{rr} \ln\left(\frac{p_{st}/p_t}{p_s/p}\right).$$

- **External Demand:**

$$Y_t^{*} = A_t^{g} y^{*},$$

$$X_t = \Delta_{yt}^{*} Y_t^{*},$$

$$\Delta_{yt}^{*} = (1 - \chi^{*})\left(\tilde{P}_{rt}^{*}\right)^{-\psi^{*}} + \chi^{*}\Delta_{y,t-1}^{*}.$$

- **Market Clearing:**

$$Z_t = \Delta_{yt} Y_t + X_t$$

$$\Delta_{yt} = (1 - \chi)\left(\frac{\tilde{p}_{rt}}{p_t}\right)^{-\psi} + \chi\left(\frac{\pi_t}{\pi_{t-1}}\right)^{\psi}\Delta_{y,t-1},$$

$$Y_t = C_t + \left[1 + \frac{\Omega_k}{2}\left(\frac{I_t}{I_{t-1}} - \exp(g)\right)^2\right] I_t.$$

- **Balance of Payment:**

$$\Delta_{pt}^{*} Y_t^{*} - M_t = R_{t-1}^{*} B_{t-1}^{*} - B_t^{*},$$

$$\Delta_{pt}^{*} = (1-\chi^{*})\left(\tilde{P}_{rt}^{*}\right)^{1-\psi^{*}} + \chi^{*}\Delta_{p,t-1}^{*}.$$

- **Exogenous Shocks:**

$$a_t = \rho_a a_{t-1} + \varepsilon_{at},$$

$$\ln r_t^{*} = (1-\rho_{r^*})\ln r^{*} + \rho_{r^*}\ln r_{t-1}^{*} + \varepsilon_{r^*t},$$

$$\ln s_t = (1-\rho_s)\ln s + \rho_s \ln s_{t-1} + \varepsilon_{st}.$$

B.2. Detrended System

- **Households:**

(B.1)
$$\tilde{\lambda}_t = \frac{1}{p_t}\left(\frac{1}{c_t - h\frac{c_{t-1}}{\exp(g)}} - \mathbb{E}_t\beta\frac{h}{c_{t+1}\exp(g) - hc_t}\right),$$

(B.2)
$$\tilde{\lambda}_t\tilde{w}_t = \kappa N_t^{\varphi}.$$

- **Banks:**

$$\tilde{l}_t + \tilde{re}_t = \tilde{b}_t + b_t^{*},$$

$$\tilde{re}_t = \frac{rr_t}{1-rr_t}\tilde{l}_t,$$

$$R_{lt} = \frac{R_t - rr_t}{1-rr_t},$$

$$\mathbb{E}_t\beta\frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t\exp(g)}\left[\frac{R_t}{\pi_{e,t+1}} - R_t^{*}\right] = \mathbb{E}_t\beta\frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t\exp(g)}\left[\Omega\exp\left(\frac{b_t^{*}}{p_{zt}z_t} - \underline{B}^{*}\right)\frac{b_t^{*}}{p_{zt}z_t}\right],$$

$$R_t^{*} = r_t^{*} + \Omega\left[\exp\left(\frac{b_t^{*}}{p_{zt}z_t} - \underline{B}^{*}\right) - 1\right].$$

- **Entrepreneurs:**

$$\bar{\varepsilon}_t = \frac{p_{kt}}{q_t},$$

$$\bar{\varepsilon}_{lt} = \frac{R_{lt}}{R_t}\bar{\varepsilon}_t,$$

$$q_t = \mathbb{E}_t \beta \frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t \exp(g)} \Bigg[r_{k,t+1} + (1-\delta) q_{t+1} + r_{k,t+1} \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) + \mu p_{k,t+1} \int_{\bar{\varepsilon}_{l,t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - \frac{R_{l,t+1}}{R_{t+1}} \right) dF(\varepsilon) \Bigg],$$

$$\frac{1}{R_t} = \mathbb{E}_t \beta \frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t \exp(g) \pi_{e,t+1}} \left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) \right],$$

(B.3) $$\tilde{p}_t^h = \mathbb{E}_t \beta \frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t} \tilde{p}_{e,t+1}^h \left[1 + \int_{\bar{\varepsilon}_{t+1}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}_{t+1}} - 1 \right) dF(\varepsilon) \right],$$

(B.4) $$\tilde{p}_{et}^h = s_t \tilde{p}_t^h,$$

$$\tilde{l}_t = \mu p_{kt} \frac{k_{t-1}}{\exp(g)} \left(1 - F(\bar{\varepsilon}_{lt}) \right),$$

$$\tilde{nw}_t = r_{kt} \frac{k_{t-1}}{\exp(g)} + \frac{1}{\pi_{et}} \frac{\tilde{re}_{t-1}}{\exp(g)} - \frac{R_{t-1}}{\pi_{et}} \frac{b_{t-1}^*}{\exp(g)} + \tilde{p}_t^h,$$

(B.5) $$i_t = \frac{1}{p_{kt}} \left[\tilde{nw}_t \left(1 - F(\bar{\varepsilon}_t) \right) + \tilde{l}_t \right],$$

(B.6) $$k_t = (1-\delta) \frac{k_{t-1}}{\exp(g)} + \left(i_t - \frac{\tilde{l}_t}{p_{kt}} \right) \frac{\int_{\bar{\varepsilon}_t}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_t)} + \frac{\tilde{l}_t}{p_{kt}} \frac{\int_{\bar{\varepsilon}_{lt}}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon)}{1 - F(\bar{\varepsilon}_{lt})},$$

$$\tilde{p}_{st} = q_t k_t + \frac{\tilde{re}_t}{R_t} - b_t^* + \tilde{p}_t^h.$$

- **Wholesale Goods Producers:**

(B.7) $$z_t = \left[\frac{k_{t-1}}{\exp(g)} \right]^{\alpha} \left[\exp(a_t) N_t \right]^{1-\alpha-\gamma} m_t^{\gamma},$$

(B.8) $$r_{kt} = \alpha \frac{p_{zt} z_t}{k_{t-1}} \exp(g),$$

(B.9) $$\tilde{w}_t = (1-\alpha-\gamma) \frac{p_{zt} z_t}{N_t},$$

(B.10) $$m_t = \gamma p_{zt} z_t.$$

- **Capital Producers:**

$$p_{kt} = \left[1 + \frac{\Omega_k}{2}\left(\frac{i_t}{i_{t-1}}\exp(g) - \exp(g)\right)^2\right] p_t + \Omega_k p_t \left(\frac{i_t}{i_{t-1}}\exp(g) - \exp(g)\right)\frac{i_t}{i_{t-1}}\exp(g)$$
$$- \Omega_k \mathbb{E}_t \beta \frac{\tilde{\lambda}_{t+1}}{\tilde{\lambda}_t \exp(g)} p_{t+1}\left(\frac{i_{t+1}}{i_t}\exp(g) - \exp(g)\right)\left(\frac{i_{t+1}}{i_t}\exp(g)\right)^2.$$

- **Retailers:**

$$\pi_t = \pi_{et}\frac{p_t}{p_{t-1}},$$

$$p_t = \left[(1-\chi)\tilde{p}_{rt}^{1-\psi} + \chi \pi_{t-1}^{1-\psi}\left(\frac{p_{t-1}}{\pi_{et}}\right)^{1-\psi}\right]^{\frac{1}{1-\psi}},$$

$$\tilde{p}_{rt} = \frac{\psi}{\psi - 1}\frac{\gamma_{nt}}{\gamma_{dt}},$$

$$\gamma_{nt} = \tilde{\lambda}_t p_{zt} p_t^{\psi} y_t + \chi\beta\pi_t^{-\psi}\mathbb{E}_t \pi_{e,t+1}^{\psi}\gamma_{n,t+1},$$

$$\gamma_{dt} = \tilde{\lambda}_t p_t^{\psi} y_t + \chi\beta\pi_t^{1-\psi}\mathbb{E}_t \pi_{e,t+1}^{\psi-1}\gamma_{d,t+1},$$

$$\tilde{P}_{rt}^* = \frac{\psi^*}{\psi^* - 1}\frac{\Gamma_{nt}^*}{\Gamma_{dt}^*},$$

$$\Gamma_{nt}^* = \tilde{\lambda}_t p_{zt} y_t^* + \chi^*\beta\mathbb{E}_t\Gamma_{n,t+1}^*,$$

$$\Gamma_{dt}^* = \tilde{\lambda}_t y_t^* + \chi^*\beta\mathbb{E}_t\Gamma_{d,t+1}^*.$$

- **Central Bank:**

$$\ln\left(\frac{R_t}{R}\right) = \phi_\pi \ln\left(\frac{\pi_t}{\pi}\right) + \phi_z \ln\left(\frac{y_t}{y}\right) + \phi_f \ln\left(\frac{\tilde{p}_{st}/p_t}{\tilde{p}_s/p}\right),$$

$$rr_t - rr = \phi_{rr}\ln\left(\frac{\tilde{p}_{st}/p_t}{\tilde{p}_s/p}\right).$$

- **External Demand:**

(B.11) $$x_t = \Delta_{yt}^* y^*,$$

(B.12) $$\Delta_{yt}^* = (1-\chi^*)\left(\tilde{P}_{rt}^*\right)^{-\psi^*} + \chi^*\Delta_{y,t-1}^*.$$

- **Market Clearing:**

(B.13)
$$z_t = \Delta_{yt} y_t + x_t,$$

$$\Delta_{yt} = (1-\chi)\left(\frac{\tilde{p}_{rt}}{p_t}\right)^{-\psi} + \chi\left(\frac{\pi_t}{\pi_{t-1}}\right)^{\psi} \Delta_{y,t-1},$$

(B.14)
$$y_t = c_t + \left[1 + \frac{\Omega_k}{2}\left(\frac{i_t}{i_{t-1}}\exp(g) - \exp(g)\right)^2\right] i_t.$$

- **Balance of Payment:**

(B.15)
$$\Delta^*_{pt} y^* - m_t = R^*_{t-1}\frac{b^*_{t-1}}{\exp(g)} - b^*_t,$$

(B.16)
$$\Delta^*_{pt} = (1-\chi^*)\left(\tilde{P}^*_{rt}\right)^{1-\psi^*} + \chi^* \Delta^*_{p,t-1}.$$

- **Exogenous Shocks:**

$$a_t = \rho_a a_{t-1} + \varepsilon_{at},$$

$$\ln r^*_t = (1-\rho_{r^*})\ln r^* + \rho_{r^*}\ln r^*_{t-1} + \varepsilon_{r^*t},$$

$$\ln s_t = (1-\rho_s)\ln s + \rho_s \ln s_{t-1} + \varepsilon_{st}.$$

Appendix C. Proofs of Propositions in Section 3

C.1. Proof of Lemma 1

PROOF. Define the auxiliary function $\mathcal{G}(z)$ for $z \in [\varepsilon_{\min}, \varepsilon_{\max}]$ as:

$$\mathcal{G}(z) \equiv 1 + \int_z^{\varepsilon_{\max}} \left(\frac{\varepsilon}{z} - 1 \right) dF(\varepsilon).$$

Taking the first derivative of $\mathcal{G}(z)$ with respect to z yields:

$$\mathcal{G}'(z) = -\frac{1}{z^2} \int_z^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon).$$

Since $\varepsilon > 0$ and the integral is strictly positive for any $z < \varepsilon_{\max}$, it follows that $\mathcal{G}'(z) < 0$. Thus, $\mathcal{G}(z)$ is a strictly decreasing, continuous function on the interval $[\varepsilon_{\min}, \varepsilon_{\max}]$.

Evaluating $\mathcal{G}(z)$ at the boundaries of the support gives:

$$\mathcal{G}(\varepsilon_{\min}) = 1 + \int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\varepsilon_{\min}} - 1 \right) dF(\varepsilon) = \frac{1}{\varepsilon_{\min}} \int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon) = \frac{\mathbb{E}[\varepsilon]}{\varepsilon_{\min}},$$
$$\mathcal{G}(\varepsilon_{\max}) = 1 + \int_{\varepsilon_{\max}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\varepsilon_{\max}} - 1 \right) dF(\varepsilon) = 1.$$

By the assumption stated in the lemma, we are given that $\frac{\mathbb{E}[\varepsilon]}{\varepsilon_{\min}} > \pi\beta^{-1}\exp(g) > 1$. Substituting our boundary evaluations into this inequality yields:

$$\mathcal{G}(\varepsilon_{\min}) > \pi\beta^{-1}\exp(g) > \mathcal{G}(\varepsilon_{\max}).$$

Because $\mathcal{G}(z)$ is continuous and strictly monotonically decreasing, the Intermediate Value Theorem guarantees that there exists a unique threshold $\underline{\varepsilon} \in (\varepsilon_{\min}, \varepsilon_{\max})$ such that $\mathcal{G}(\underline{\varepsilon}) = \pi\beta^{-1}\exp(g)$, and this completes the proof. □

C.2. Proof of Lemma 2

PROOF. Recall the expression for the steady-state real rental rate of capital from equation (60):

$$\frac{r_k}{p_k}(\bar{\varepsilon}) = \frac{\exp(g) - \beta(1-\delta) - \mu\beta \int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}} (\varepsilon - \varepsilon_l(\bar{\varepsilon})) dF(\varepsilon)}{\beta \left[\bar{\varepsilon} + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}} (\varepsilon - \bar{\varepsilon}) dF(\varepsilon) \right]}.$$

Let the denominator be denoted as $Denom(\bar{\varepsilon}) = \beta \left[\bar{\varepsilon} + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}} (\varepsilon - \bar{\varepsilon}) dF(\varepsilon) \right]$. Applying Leibniz's integral rule, the derivative of the denominator with respect to $\bar{\varepsilon}$ is simply $\beta F(\bar{\varepsilon})$.

Taking the first derivative of the entire quotient $\frac{r_k}{p_k}(\bar{\varepsilon})$ with respect to $\bar{\varepsilon}$ using the quotient rule

yields:

$$\frac{\partial}{\partial \bar{\varepsilon}}\left(\frac{r_k}{p_k}(\bar{\varepsilon})\right) = \frac{1}{[Denom(\bar{\varepsilon})]^2}\Bigg\{\left[\mu\beta\int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}} dF(\varepsilon)\cdot\varepsilon_l'(\bar{\varepsilon})\right]Denom(\bar{\varepsilon}) - \left[\exp(g)-\beta(1-\delta)-\mu\beta\int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}}(\varepsilon-\varepsilon_l(\bar{\varepsilon}))dF(\varepsilon)\right]\beta F(\bar{\varepsilon})\Bigg\}.$$

To evaluate the sign of this complex derivative, we examine the limiting case where the pledgeability parameter (the degree of financial friction) approaches zero, i.e., $\mu = 0$. Substituting $\mu = 0$ into the derivative expression significantly simplifies the equation to:

$$\frac{\partial}{\partial \bar{\varepsilon}}\left(\frac{r_k}{p_k}(\bar{\varepsilon})\right)\Bigg|_{\mu=0} = \frac{-[\exp(g)-\beta(1-\delta)]\beta F(\bar{\varepsilon})}{\beta^2\left[\bar{\varepsilon}+\int_{\bar{\varepsilon}}^{\varepsilon_{\max}}(\varepsilon-\bar{\varepsilon})dF(\varepsilon)\right]^2}.$$

In order for the steady-state capital rental rate to be positive in a frictionless framework, it must hold that $\exp(g) > \beta(1-\delta)$. Furthermore, since $\bar{\varepsilon}$ is an interior solution such that $\bar{\varepsilon} > \varepsilon_{\min}$, the cumulative distribution function is strictly positive, $F(\bar{\varepsilon}) > 0$. Consequently, the numerator is strictly negative while the denominator is strictly positive, yielding:

$$\frac{\partial}{\partial \bar{\varepsilon}}\left(\frac{r_k}{p_k}(\bar{\varepsilon})\right)\Bigg|_{\mu=0} < 0.$$

Because the derivative $\frac{\partial}{\partial \bar{\varepsilon}}\left(\frac{r_k}{p_k}(\bar{\varepsilon})\right)$ is continuous with respect to the parameter μ, the strict inequality is preserved in the neighborhood of zero. Therefore, for a sufficiently small $\mu > 0$, $\frac{r_k}{p_k}(\bar{\varepsilon})$ is strictly decreasing in $\bar{\varepsilon}$, and this completes the proof. □

C.3. Proof of Proposition 3

PROOF. By setting $\tilde{p}^h = \tilde{p}_e^h = 0$ in equation (B.3) and (B.4), we solve for the bubbleless steady state through the following steps.

We first solve for the investment efficiency threshold $\bar{\varepsilon}$. Dividing both sides of the law of motion for aggregate capital (B.6) by $\frac{k}{\exp(g)}$ yields

$$\exp(g)-(1-\delta) = \frac{1}{p_k}\left(r_k+\frac{1}{\pi}\frac{\tilde{re}}{k}-\frac{R}{\pi}\frac{b^*}{k}\right)\int_{\bar{\varepsilon}}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon)+\mu\int_{\bar{\varepsilon}_l}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon)$$

$$= \left[\frac{r_k}{p_k}(\bar{\varepsilon})+\frac{1}{\pi}\frac{rr}{1-rr}\frac{\mu}{\exp(g)}\big(1-F(\varepsilon_l(\bar{\varepsilon}))\big) - \frac{R(\bar{\varepsilon})}{\pi}\frac{b^*}{p_z z}(\bar{\varepsilon})\frac{p_z z}{p_k k}\right]\int_{\bar{\varepsilon}}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon)+\mu\int_{\varepsilon_l(\bar{\varepsilon})}^{\varepsilon_{\max}}\varepsilon\, dF(\varepsilon).$$

Substituting the capital rent condition (B.8) into the equation above yields equation (64), where

$\mathcal{H}(\bar{\varepsilon})$ is defined by equation (62). Under Lemma 2, $\mathcal{H}(\cdot)$ is a strictly decreasing continuous function. Therefore, assumption (63), combined with the fact that $\mathcal{H}(\varepsilon_{\max}) = 0 < \exp(g) - (1-\delta)$, guarantees a unique solution $\bar{\varepsilon}_f$ to equation (64). Once $\bar{\varepsilon}_f$ is obtained, the steady-state values $R_f = R(\bar{\varepsilon}_f)$, $\bar{\varepsilon}_{lf} = \varepsilon_l(\bar{\varepsilon}_f)$, $\frac{r_{kf}}{p_{kf}} = \frac{r_k}{p_k}(\bar{\varepsilon}_f)$, and $\frac{b_f^*}{p_{zf} z_f} = \frac{b^*}{p_z z}(\bar{\varepsilon}_f)$ can be determined from equations (57), (59), (60), and (61), respectively.

The balance of payments condition (B.15) implies

$$\Delta_p^* y^* - m = b^* \left[\frac{r^*}{\exp(g)} - 1 \right].$$

From equations (B.11), (B.12), and (B.16), we know that $\Delta_p^* y^* = x \frac{\Delta_p^*}{\Delta_y^*} = \tilde{P}_r^* x = \frac{\psi^*}{\psi^* - 1} p_z x$. Substituting this and the imported input demand condition (B.10) into the equation above yields the investment-output ratio as a function $\bar{\varepsilon}$

$$\frac{x}{z} = \frac{\psi^* - 1}{\psi^*} \left[\left(\frac{r^*}{\exp(g)} - 1 \right) \frac{b^*}{p_z z}(\bar{\varepsilon}) + \gamma \right], \tag{C.1}$$

Under the second half of assumption (65), we have

$$\frac{x_f}{z_f} = \frac{\psi^* - 1}{\psi^*} \left[\left(\frac{r^*}{\exp(g)} - 1 \right) \frac{b^*}{p_z z}(\bar{\varepsilon}_f) + \gamma \right] > 0. \tag{C.2}$$

Dividing both sides of the aggregate investment condition (B.5) by z yields

$$\begin{aligned} \frac{i}{z} = \frac{1}{\exp(g)} \Bigg[& \frac{r_k}{p_k}(\bar{\varepsilon}) \frac{k}{z} + \frac{1}{\pi} \frac{rr}{1 - rr} \frac{\mu}{\exp(g)} \frac{k}{z} \big(1 - F(\varepsilon_l(\bar{\varepsilon}))\big) \\ & - \frac{R(\bar{\varepsilon})}{\pi} \frac{b^*}{p_z z}(\bar{\varepsilon}) \frac{p_z}{p_k} \Bigg] \big(1 - F(\bar{\varepsilon})\big) + \frac{\mu}{\exp(g)} \frac{k}{z} \big(1 - F(\varepsilon_l(\bar{\varepsilon}))\big). \end{aligned}$$

Substituting the capital rent condition (B.8) and $\frac{p_z}{p_k} = \frac{\psi - 1}{\psi}$ into the equation above yields the investment-output ratio as a function of $\bar{\varepsilon}$ as

$$\begin{aligned} \frac{i}{z}(\bar{\varepsilon}) = \frac{\psi - 1}{\psi} \Bigg\{ & \alpha + \frac{1}{\pi} \frac{rr}{1 - rr} \frac{\mu\alpha}{\exp(g)} \left[\frac{r_k}{p_k}(\bar{\varepsilon}) \right]^{-1} \big(1 - F(\varepsilon_l(\bar{\varepsilon}))\big) \\ & - \frac{R(\bar{\varepsilon})}{\pi} \frac{1}{\exp(g)} \frac{b^*}{p_z z}(\bar{\varepsilon}) + \mu\alpha \left[\frac{r_k}{p_k}(\bar{\varepsilon}) \right]^{-1} \frac{1 - F(\varepsilon_l(\bar{\varepsilon}))}{1 - F(\bar{\varepsilon})} \Bigg\} \big(1 - F(\bar{\varepsilon})\big), \end{aligned}$$

and in the bubbleless steady state,

$$\frac{i_f}{z_f} = \frac{\psi - 1}{\psi}\Bigg\{\alpha + \frac{1}{\pi}\frac{rr}{1 - rr}\frac{\mu\alpha}{\exp(g)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\left(1 - F(\varepsilon_l(\bar{\varepsilon}_f))\right) - \frac{R(\bar{\varepsilon}_f)}{\pi}\frac{1}{\exp(g)}\frac{b^*}{p_z z}(\bar{\varepsilon}_f) + \mu\alpha\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\frac{1 - F(\varepsilon_l(\bar{\varepsilon}_f))}{1 - F(\bar{\varepsilon}_f)}\Bigg\}\left(1 - F(\bar{\varepsilon}_f)\right). \tag{C.3}$$

From the resource constraints (B.13) and (B.14), we have

$$c = y - i = z - x - i,$$

which implies that the consumption-output ratio is given by

$$\begin{aligned}\frac{c_f}{z_f} &= 1 - \frac{x_f}{z_f} - \frac{i_f}{z_f} \\ &= 1 - \frac{\psi^* - 1}{\psi^*}\left[\left(\frac{r^*}{\exp(g)} - 1\right)\frac{b^*}{p_z z}(\bar{\varepsilon}_f) + \gamma\right] \\ &\quad - \frac{\psi - 1}{\psi}\Bigg\{\alpha + \frac{1}{\pi}\frac{rr}{1 - rr}\frac{\mu\alpha}{\exp(g)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\left(1 - F(\varepsilon_l(\bar{\varepsilon}_f))\right) \\ &\quad - \frac{R(\bar{\varepsilon}_f)}{\pi}\frac{1}{\exp(g)}\frac{b^*}{p_z z}(\bar{\varepsilon}_f) + \mu\alpha\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\frac{1 - F(\varepsilon_l(\bar{\varepsilon}_f))}{1 - F(\bar{\varepsilon}_f)}\Bigg\}\left(1 - F(\bar{\varepsilon}_f)\right)\end{aligned} \tag{C.4}$$

which is also positive under the first half of assumption (65).

Combining the labor supply condition (B.2), the labor demand condition (B.9), and the marginal utility of consumption (B.1) yields

$$\kappa N^{1+\varphi} = (1 - \alpha - \gamma)\frac{\psi - 1}{\psi}\frac{\exp(g) - \beta h}{\exp(g) - h}\frac{z}{c},$$

from which we can pin down working hours at the bubbleless steady state as

$$N_f = \left[\frac{1 - \alpha - \gamma}{\kappa}\frac{\psi - 1}{\psi}\frac{\exp(g) - \beta h}{\exp(g) - h}\left(\frac{c_f}{z_f}\right)^{-1}\right]^{\frac{1}{1+\varphi}}.$$

From the production technology (B.7), we obtain

$$z = \left[\frac{1}{\exp(g)}\frac{k}{z}\right]^{\frac{\alpha}{1-\alpha-\gamma}}\left(\frac{m}{z}\right)^{\frac{\gamma}{1-\alpha-\gamma}} N.$$

Plugging in the factor demands (B.8) and (B.10) yields

$$z = \left(\alpha \frac{\psi - 1}{\psi} \left[\frac{r_k}{p_k}(\bar{\varepsilon})\right]^{-1}\right)^{\frac{\alpha}{1-\alpha-\gamma}} (\gamma p_z)^{\frac{\gamma}{1-\alpha-\gamma}} N. \tag{C.5}$$

Combining this with equations (B.11) and (B.12) yields

$$\frac{\left(\frac{\psi^*}{\psi^*-1} p_z\right)^{-\psi^*} y^*}{\left(\alpha \frac{\psi-1}{\psi}\left[\frac{r_k}{p_k}(\bar{\varepsilon})\right]^{-1}\right)^{\frac{\alpha}{1-\alpha-\gamma}} (\gamma p_z)^{\frac{\gamma}{1-\alpha-\gamma}} N} = \frac{x}{z},$$

from which we can solve for the wholesale goods price at the bubbleless steady state as

$$p_{zf} = \frac{\left(\frac{\psi^*}{\psi^*-1}\right)^{-\frac{\psi^*(1-\alpha-\gamma)}{\psi^*(1-\alpha-\gamma)+\gamma}} y^{*\frac{1-\alpha-\gamma}{\psi^*(1-\alpha-\gamma)+\gamma}}}{\left(\frac{x_f}{z_f}\right)^{\frac{1-\alpha-\gamma}{\psi^*(1-\alpha-\gamma)+\gamma}} \left(\alpha \frac{\psi-1}{\psi}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_f)\right]^{-1}\right)^{\frac{\alpha}{\psi^*(1-\alpha-\gamma)+\gamma}} \gamma^{\frac{\gamma}{\psi^*(1-\alpha-\gamma)+\gamma}} N_f^{\frac{1-\alpha-\gamma}{\psi^*(1-\alpha-\gamma)+\gamma}}}.$$

Substituting this result into equation (C.5) yields z_f. Plugging $p_{kf} = \frac{\psi}{\psi-1} p_{zf}$ into equation (60) gives r_{kf}. Given z_f, equations (61), (C.2), (C.3), and (C.4) determine the levels b_f^*, x_f, i_f, and c_f, respectively. All remaining variables can be easily determined by referring to the full detrended equilibrium system outlined in Appendix B.2. This completes the proof. □

C.4. Proof of Proposition 4

PROOF. We first prove the necessity and sufficiency of the bubble existence condition (66).

We begin with necessity. Suppose $\tilde{p}^h > 0$ and let $\bar{\varepsilon}_b$ be a solution to equation (67). Dividing both sides of the law of motion for aggregate capital (B.6) by $\frac{k}{\exp(g)}$ yields

$$\exp(g) - (1-\delta) = \mathcal{H}(\bar{\varepsilon}) + \frac{\tilde{p}^h}{p_k k} \exp(g) \int_{\bar{\varepsilon}}^{\varepsilon_{\max}} \varepsilon \, dF(\varepsilon), \tag{C.6}$$

where $\mathcal{H}(\bar{\varepsilon})$ is given by equation (62).

Since $\mathcal{H}(\cdot)$ is a strictly decreasing function and the bubbleless steady state with $\tilde{p}^h = 0$ is characterized by $\exp(g) - (1-\delta) = \mathcal{H}(\bar{\varepsilon}_f)$, the assumption $\tilde{p}^h > 0$ dictates that $\mathcal{H}(\bar{\varepsilon}_b) < \mathcal{H}(\bar{\varepsilon}_f)$, which then implies $\bar{\varepsilon}_b > \bar{\varepsilon}_f$.

Now, define the left-hand side of equation (67) as the function $\mathcal{G}(\bar{\varepsilon})$:

$$\mathcal{G}(\bar{\varepsilon}) \equiv s\beta \left[1 + \int_{\bar{\varepsilon}}^{\varepsilon_{\max}} \left(\frac{\varepsilon}{\bar{\varepsilon}} - 1\right) dF(\varepsilon)\right].$$

The function $\mathcal{G}(\bar{\varepsilon})$ is strictly decreasing in $\bar{\varepsilon}$. Therefore, $\bar{\varepsilon}_b > \bar{\varepsilon}_f$ implies $\mathcal{G}(\bar{\varepsilon}_f) > \mathcal{G}(\bar{\varepsilon}_b) = 1$, which directly yields the existence condition (66).

Conversely, to prove sufficiency, suppose the bubble existence condition (66) holds such that $\mathcal{G}(\bar{\varepsilon}_f) > 1$. Since $\mathcal{G}(\cdot)$ is a continuous, strictly decreasing function with the boundary property $\mathcal{G}(\varepsilon_{\max}) = 0$, there must exist a unique $\bar{\varepsilon}_b > \bar{\varepsilon}_f$ such that $\mathcal{G}(\bar{\varepsilon}_b) = 1$ and hence satisfies the steady state bubble pricing condition (B.3).

The bubbleless steady state features $\mathcal{H}(\bar{\varepsilon}_f) = \exp(g) - (1-\delta)$. As $\bar{\varepsilon}_b > \bar{\varepsilon}_f$ and $\mathcal{H}(\cdot)$ is strictly decreasing, it follows that $\mathcal{H}(\bar{\varepsilon}_b) < \exp(g) - (1-\delta)$. Substituting this inequality back into the law of motion for aggregate capital (C.6) immediately yields $\tilde{p}^h > 0$, so that a bubbly steady state exists.

Once $\bar{\varepsilon}_b$ is obtained, the steady-state values $R_b = R(\bar{\varepsilon}_b)$, $\bar{\varepsilon}_{lb} = \varepsilon_l(\bar{\varepsilon}_b)$, $\frac{r_{kb}}{p_{kb}} = \frac{r_k}{p_k}(\bar{\varepsilon}_b)$, and $\frac{b^*_b}{p_{zb}z_b} = \frac{b^*}{p_z z}(\bar{\varepsilon}_b)$ can be easily determined.

Following equation (B.5), the investment-output ratio with $\tilde{p}^h > 0$ becomes

$$\frac{i_b}{z_b} = \frac{\psi - 1}{\psi}\Bigg\{\alpha + \frac{1}{\pi}\frac{rr}{1-rr}\frac{\mu\alpha}{\exp(g)}\left[\frac{r_k}{p_k}(\bar{\varepsilon}_b)\right]^{-1}\left(1 - F(\varepsilon_l(\bar{\varepsilon}_b))\right)$$

$$- \frac{R(\bar{\varepsilon}_b)}{\pi}\frac{1}{\exp(g)}\frac{b^*}{p_z z}(\bar{\varepsilon}_b) + \mu\alpha\left[\frac{r_k}{p_k}(\bar{\varepsilon}_b)\right]^{-1}\frac{1 - F(\varepsilon_l(\bar{\varepsilon}_b))}{1 - F(\bar{\varepsilon}_b)} + \frac{\tilde{p}^h_b}{p_{zb}z_b}\Bigg\}\left(1 - F(\bar{\varepsilon}_b)\right),$$

where the bubble-to-output ratio $\frac{\tilde{p}^h_b}{p_{zb}z_b}$ is given by equation (69), which is derived from equation (C.6) above. Because the export-output ratio schedule, $\frac{x}{z}(\bar{\varepsilon})$, remains the same as in the bubbleless steady state given by equation (C.1), the condition that guarantees $x_b > 0$ and $c_b > 0$ becomes (68).

The procedure for solving the steady-state values of the remaining variables is identical to that of the bubbleless equilibrium, and is hence omitted here. This completes the proof. □